NED University of Engineering and Technology
Department of Software Engineering Batch-2024

GROUP 9
LAB WORKBOOK

Course Title: Software Design and Architecture (Practical)
Course Code: SE-308
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Section: A
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SDA Lab 1

1. List the advantages of Unified Modeling Language.

- **Standardization:** UML provides a common, standardized way of modeling systems so all stakeholders can understand.
- **Visualization:** Complex systems are easier to understand when represented in diagrams instead of plain text.
- **Communication:** Acts as a bridge between developers, designers, business users, and clients.
- **Documentation:** UML diagrams serve as permanent documentation for systems, useful for future maintenance.
- **Design Validation:** Helps validate system requirements and design before coding begins, reducing errors.
- **Reusability:** Encourages modular design, making system components reusable in other projects.

2.Explain different kinds of UML diagram and their significance.

Use Case Diagram:

Use case diagram represents the functionality of a system from the user’s point of view. It shows different actors (users or external systems) and the interactions they have with the system to achieve specific goals.

Class Diagram:

Class diagram is the backbone of object-oriented modeling. It shows the classes, their attributes, operations, and relationships, providing a static view of the system’s structure.

Object Diagram:

Object diagram is like a snapshot of the system at a specific moment in time. It shows objects, their states, and the links between them, helping in understanding real examples of class relationships.

Activity Diagram:

Activity diagram models workflows or business processes. It shows the flow of control from one activity to another, making it useful for analyzing processes or system operations.

Sequence Diagram:

Sequence diagram focuses on how objects interact over time. It shows the sequence of messages exchanged between objects to accomplish a task, emphasizing time order.

State Machine Diagram:

State machine diagram models the different states of an object during its lifetime. It shows how an object transitions from one state to another in response to events.

Component Diagram:

Component diagram shows the physical modules or components of a system. It describes how these components are connected and interact to build the system.

Deployment Diagram:

Deployment diagram represents the physical architecture of the system. It shows how software components are deployed across hardware nodes, useful for distributed systems.

Package Diagram:

Package diagram helps organize UML elements into groups. It simplifies complex models by dividing them into smaller, more manageable packages.

3. Explain Some possible challenges of UML in software development in your own words.

- Gap between diagrams and implementation: UML diagrams describe the design but do not directly translate into code, so developers must still interpret them.
- Ambiguity in interpretation: Different stakeholders may interpret the same diagram differently, leading to misunderstandings.
- Complexity of large systems: Modeling very large systems may require many diagrams, which can become overwhelming and difficult to manage.
- Tool limitations: Not all UML tools integrate smoothly with modern programming environments, making them less practical.
- Maintenance effort: Updating diagrams to reflect system changes can be time-consuming and often neglected.
- Communication barriers: Non-technical stakeholders may still find some UML diagrams difficult to understand.

4. State some real -world Applications of UML.

- **Software Development**: Used in designing applications, databases, and web systems.
- **Business Process Modeling**: Visualizing workflows, organizational processes, and decision- making flows.
- **Embedded Systems**: Modeling hardware-software interactions in devices (e.g., ATMs, printers, medical devices).
- **Telecommunication Systems**: Designing network protocols and communication flows.
- **Banking and Finance**: Modeling transactions, customer management, and fraud detection systems.
- **E-commerce Systems**: Designing order processing, payment systems, and inventory management.

5.State 4+1 view Model and why it is called 4+1 instead of 5 View model.

The 4+1 View Model describes software architecture using four main views logical, process, physical, and development along with one additional view called the scenario or use case view. The four primary views represent the system from different technical perspectives, while the scenario view ties them together by showing real-world use cases that validate the design. It is called 4+1 instead of 5 because the scenario view is not an independent perspective but rather a complementary one that integrates and verifies the other four.

SDA Lab 2

1.Explain 4+1 View Model briefly.

1. Logical View
Description: Logical view is concerned with the system's functionality from the user's perspective i.e. the functional requirements, focusing on how major classes, objects, and their relationships support requirements.

Example: In online banking, classes like Customer, Account, and Transaction.
Stakeholders: End users, Analysts, Designers.
Diagrams Used : Class diagram, Object diagram, Package diagram.

2. Process View
Description: Process view is concerned with the dynamic (runtime) behavior, concurrency, and communication between processes or threads.

Example: In online banking, a Transaction Service working in parallel with a Notification Service.
Stakeholders: System integrators, Developers.
Diagrams Used : Component diagram, package diagram.

3. Development View (Implementation View)
Description: Development view is concerned with the organization of software in the development environment, focusing on modules, components, and configuration management.

Example: In an online banking system, the code is divided into modules like Authentication Service, Transaction Service etc.
Stakeholders: Programmers, Project managers.
Diagrams Used: Sequence diagram, Activity diagram, State machine diagram.

4. Physical View (Deployment View)
Description: Physical view is concerned with the mapping of software onto hardware nodes (servers, networks, devices).

Example: In an online banking system the application is deployed with a web server, an application server and a database server on separate machines.
Stakeholders: System engineers, Network engineers.
Diagrams Used : Deployment Diagram.

5. Scenarios (+1 View)
Description: Scenario view is concerned with use cases that validate the system architecture by showing how all views work together.

Example: In online banking, a user logs in, checks balance, and transfers money.
Stakeholders: All stakeholders i.e. Programmers, Managers, network engineers, end users etc.
Diagrams Used: Use Case diagram, Sequence diagram.

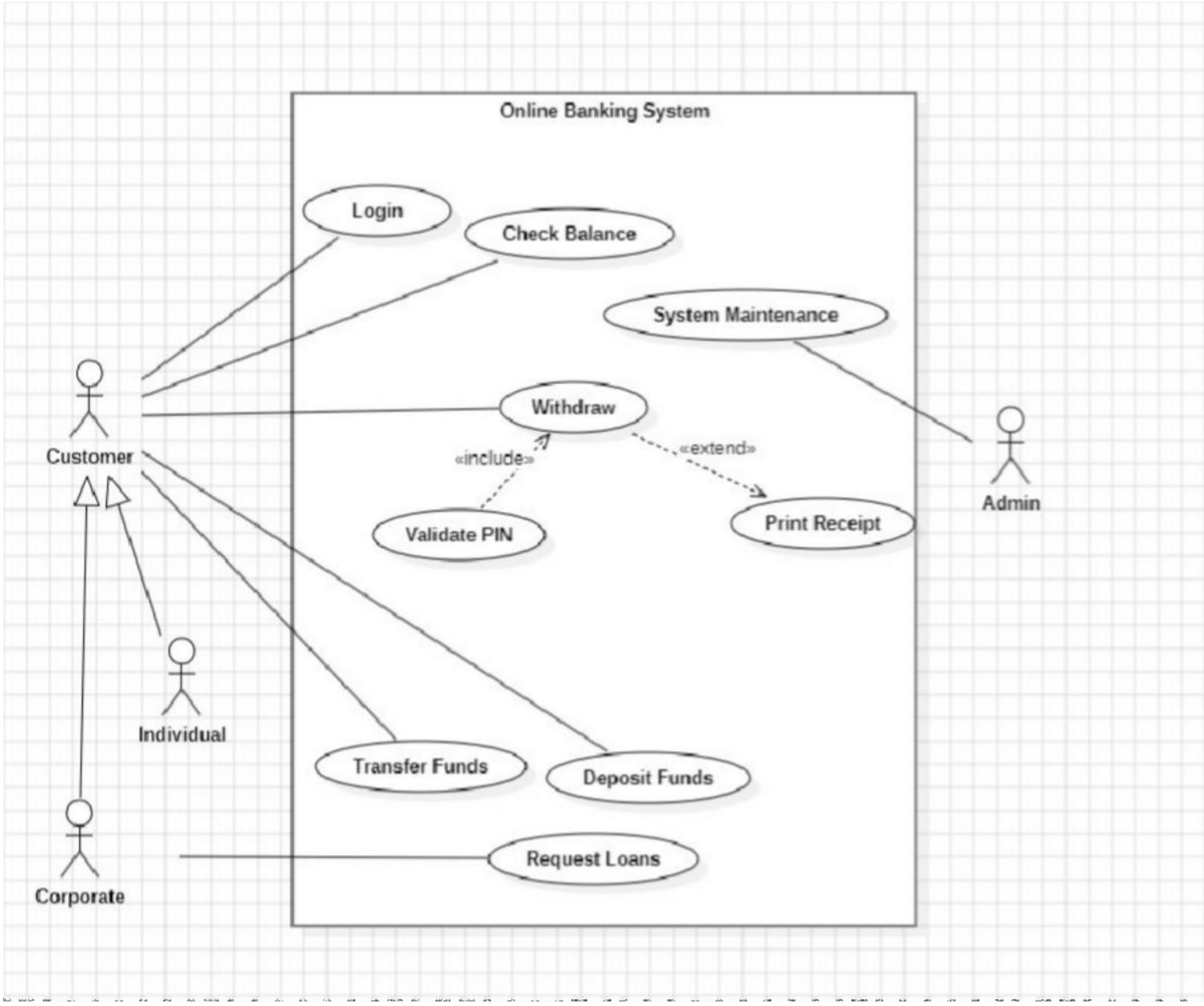
2. Install StarUML & state its key features.

StarUML is a software modeling tool used for creating and managing UML (Unified Modeling Language) diagrams. It helps in designing, visualizing, and documenting software systems efficiently.

Key Features:

- Supports creation of all major UML diagrams (class, sequence, activity, use case, component, deployment, etc.).
- Provides code generation for languages like Java, C++, C#, Python.
- Offers reverse engineering to generate UML diagrams from existing code.
- Supports multiple diagram views of the same system for better understanding.
- Exports diagrams and models into formats like HTML, PDF, RTF, and images
- Supports collaboration by sharing models with team members.
- Runs on Windows, Linux, and macOS.

3. Create a dummy project and insert any UML diagram and set properties through Editor Holders and save it.



4. Compare StarUML with any other similar tool.

Feature	StarUML	Visual Paradigm
Ease of Use	Simple, lightweight, easy to use.	More complex, harder to learn.
Functions	UML diagrams, code generation, reverse engineering.	UML, ERD, requirement modelling , stronger features.
Collaboration	Limited team support	Strong team collaboration, real-time editing
Performance	Works well for small/medium sized projects.	Better for large/enterprise projects.
Cost	Cheaper, Student-friendly pricing.	More expensive, enterprise level cost.

SDA Lab 3

1. Draw a Use-Case diagram for an Online Shopping System.
Hint: Web Customer actor uses some web site to make purchases online. Use cases are View Items, Make Purchase and Client Register. View Items use case could be used by customer if he only wants to find and see some products. This use case could also be used as a part of Make Purchase usecase. Client Register use case allows customer to register on the web site through <<Service>>Authentication actor. Note, that Checkout use case is included use case not available by itself - checkout is part of making purchase. View Items use case is extended by several optional usecases - customer may search for items and browse catalog .Checkout use case includes Customer Authentication. It could be done through user login page (sign in), Remember me page. Actor: Service Authentication Checkout use case also includes Payment use case which could be done either by using credit card and external credit payment service or with PayPal.(Actors: Credit Payment Service, PayPal. (attach Printout)

<u>Assumptions:</u>

1) The system is always online and available to users.
2) Product catalog and inventory are up-to-date.
3) Payment gateways are reliable and secure and always working.
4) System sends notifications (email/SMS) for successful orders.
5) Discounts, offers, and taxes are applied automatically based on system rules.
6) Sensitive customer data is encrypted and securely stored.
7) All External Services are Available all the time.

Diagram:

Online Shopping System
View Items
Search Items
Browse Catalog
«include»
Send Notification
Checkout
«include»
«extend»
Festival discount
Make Purchase
«include»
Payment
Notification Service
View Order History
COD
E-Wallet
Product Review
Customer
Payment Service
Track Shipment
Add to Cart
Request Item Refund
Shipping Service
Subscribe to Newletter
Chat with Customer Support
Registered Customer
New Customer
Client Register

2. A city-based ride-hailing platform allows passengers to request rides using a mobile app. Drivers can accept or reject ride requests, navigate to pick-up points, and drop passengers at their destinations. The system supports real-time tracking, fare estimation, and multiple payment options (cash, credit card, wallet). Admins manage accounts, handle complaints, and monitor ride history. A payment gateway processes all digital payments.

a. Identify Actors – List all users and external systems interacting with the application.

b. Identify Use Cases – List the main actions each actor can perform.

c. Draw the Use Case Diagram in StarUML(attach Printout)

d. Show the system boundary as Ride-Hailing App.

e. Place actors outside and use cases inside.

f. Use include and extend relationships where applicable.

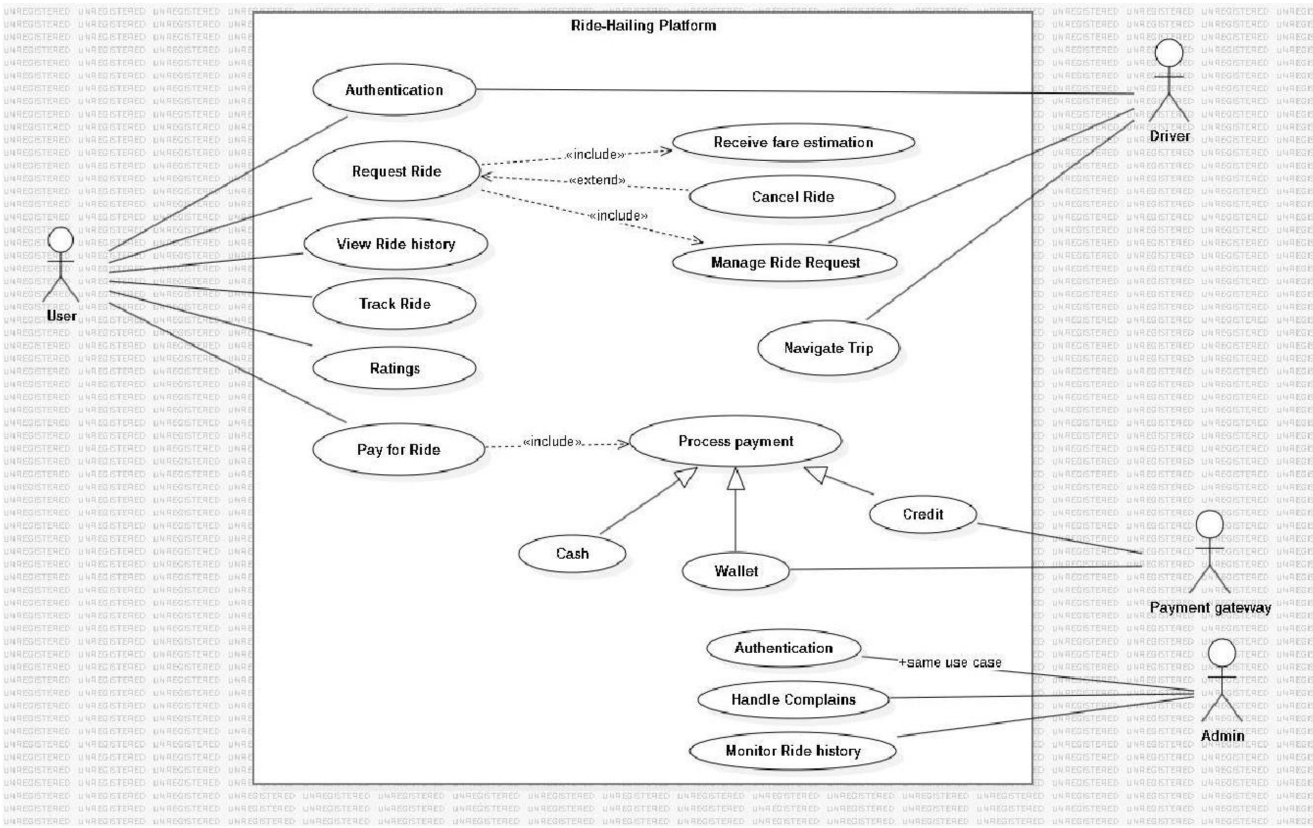
g. Export your diagram as PNG and submit both the .mdj file and image file.

Hint: Consider features like ride booking, cancellation, ride tracking, payments, and ratings.

<u>Assumptions:</u>

1) The system is always online and can handle ride requests in real-time.
2) GPS and mapping services are available and accurate.
3) Payment processing is reliable and secure.
4) Notifications (SMS/app push) for ride status updates are sent promptly.
5) Surge pricing, discounts, and fare calculations are applied automatically based on system rules.
6) Driver is shown as an actor on the right side of the system rather than the left side conventionally as it was overlapping the user's usecases.
7) Cash Payment is not connected with the external payment gateway service.

Diagram:



Ride-Hailing Platform
Authentication
Request Ride
View Ride history
Track Ride
Ratings
Pay for Ride
Receive fare estimation
Cancel Ride
Manage Ride Request
Navigate Trip
«include»
«extend»
«include»
«include»
Process payment
Cash
Wallet
Credit
Authentication
+same use case
Handle Complains
Monitor Ride history
User
Driver
Payment gateway
Admin

SDA Lab 4

1. Draw a class diagram for Retail Catalog Order.

Hint: The central class is the Order. Associated with it is the Customer making the purchase and the Payment. A Payment is one of three kinds: Cash, Check, or Credit. The Order contains OrderDetails, each with its associated Item. Attached Printout here Note: Make necessary assumptions and Show relationship among classes.

Assumptions:

1) One customer can place many orders (1 → 0..*).

2) Each order belongs to exactly one customer (many → 1).

3) One order must have at least one order detail (1 → 1..*).

4) Each order detail belongs to exactly one order (many → 1).

5) Many items can have one order detail (* - 1).

6) One item can appear in many order details (1 → 0..*).

7) One order can have zero or one payment (1 → 0..1).

8) Each payment belongs to exactly one order (1 → 1).

9) Payment is specialized into exactly one type: Cash, Check, or Credit.

10) A customer can make many payments, but only through their orders.

11) A customer can exist in the system without placing an order (just registered).

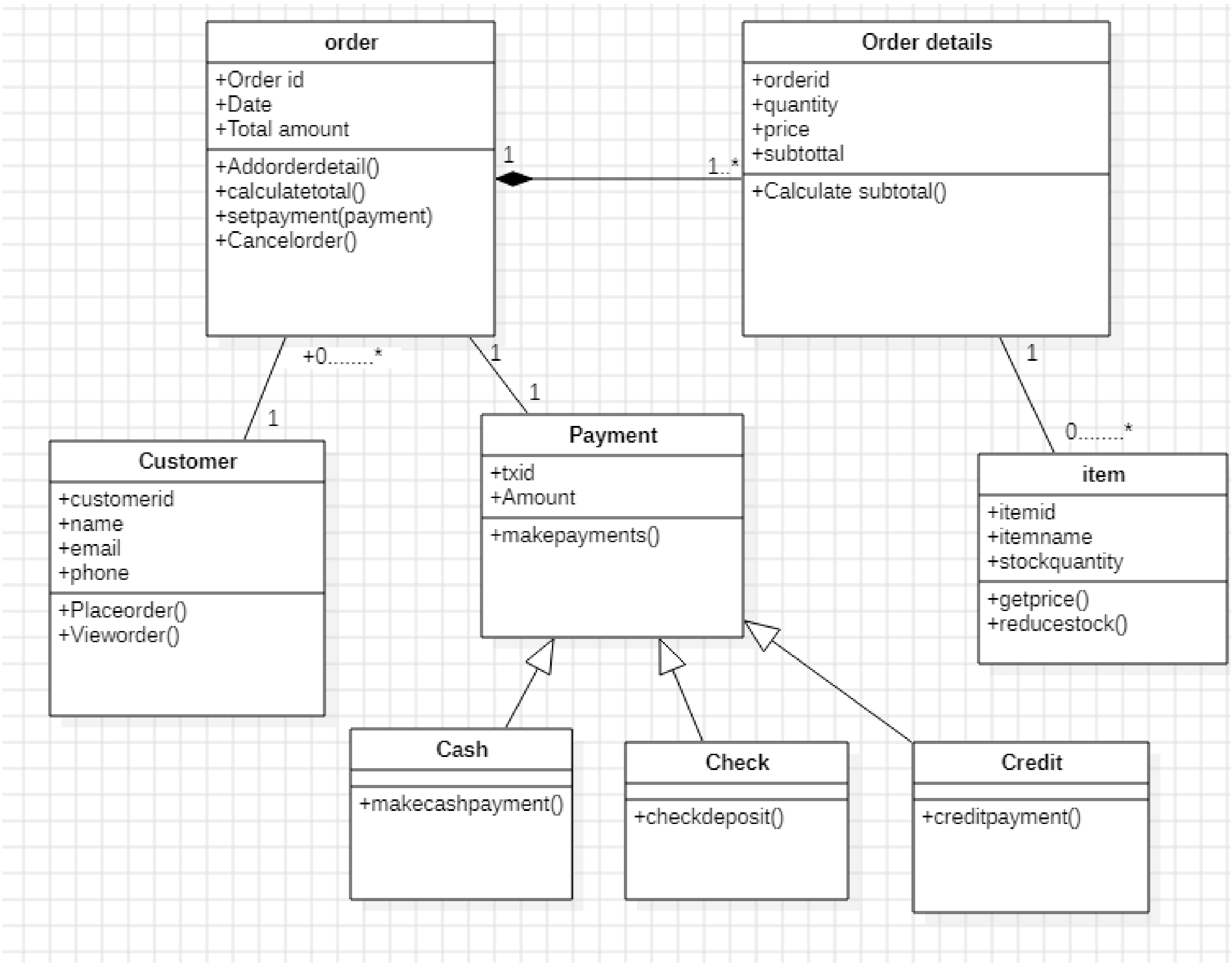
12) Order details cannot exist independently; they only exist as part of an order.

13) Each transaction ID (txid) in payment is unique.

14) Customer contact info (email/phone) is assumed to be unique per customer.

15) A payment cannot be reused for multiple orders.

Diagram:



order
+Order id
+Date
+Total amount
+Addorderdetail()
+calculatetotal()
+setpayment(payment)
+Cancelorder()
Order details
+orderid
+quantity
+price
+subtottal
+Calculate subtotal()
1
1..*
+0........*
1
1
1
1
0........*
Customer
+customerid
+name
+email
+phone
+Placeorder()
+Vieworder()
Payment
+txid
+Amount
+makepayments()
item
+itemid
+itemname
+stockquantity
+getprice()
+reducestock()
Cash
+makecashpayment()
Check
+checkdeposit()
Credit
+creditpayment()

2. Draw class diagram of Library Management system.

Hint: classes are Library, Book, Journal and Members. The central class is library, Associated with Members who can issue books or journals depending upon one of two types of member's student or staff. Staff has privilege over student to issue journal(s) as well while student can only issue book(s). Library is also associated to Book and Journal in such a way that each book/journal only associated to exactly one library. Staff can issue maximum of 12 books per semester but students can issue maximum of 6 books per semester. Attached Printout here

<u>Assumptions:</u>

1) One library can have many members (1 → *).

2) One library can have many books (1 → *).

3) One library can have many journals (1 → *).

4) Each member belongs to exactly one library (* → 1).

5) Member can issue books or journals (depending on two types of member's student or staff).

6) A student (subclass of Member) can issue up to 6 books at a time (1 → 0..6).

7) A staff (subclass of Member) can issue up to 12 books at a time (1 → 0..12).

8) A staff member can issue many journals (1 → *).

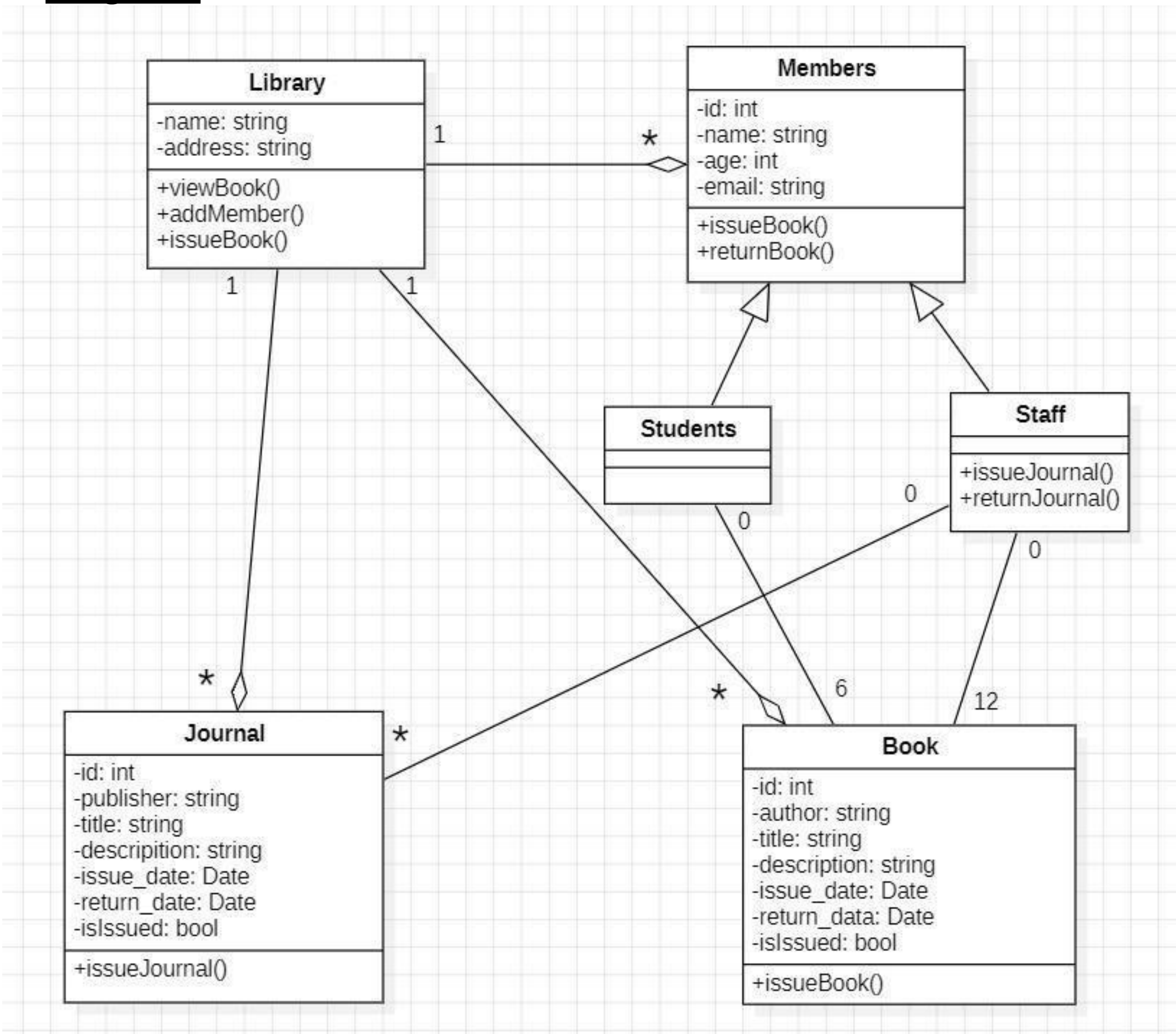
9) A journal can be issued to only one staff at a time (* → 0..1).

10) A student cannot issue journals (only staff can).

11) Each book belongs to exactly one library (* → 1).

12) Each journal belongs to exactly one library (* → 1).

Diagram:


Library
-name: string
-address: string
+viewBook()
+addMember()
+issueBook()
Members
-id: int
-name: string
-age: int
-email: string
+issueBook()
+returnBook()
Students
Staff
+issueJournal()
+returnJournal()
Journal
-id: int
-publisher: string
-title: string
-descripition: string
-issue_date: Date
-return_date: Date
-isIssued: bool
+issueJournal()
Book
-id: int
-author: string
-title: string
-description: string
-issue_date: Date
-return_data: Date
-isIssued: bool
+issueBook()
1
*
1
1
0
0
0
*
*
*
6
12

SDA LAB 5

1. Draw State Diagram for Airline Reservation System. Show different activities involved for making Reservation, Changing Reservation and Passenger Checking in for flight. Also clearly write down your assumptions.

Assumptions:

1) The user interacts through a web based interface that sends events like Flight Found, Payment Success etc.

2) The user performs each step sequentially and they cannot skip directly to payment or check-in without going through previous states.

3) The system already has access to flight schedules and seat availability through an external database.

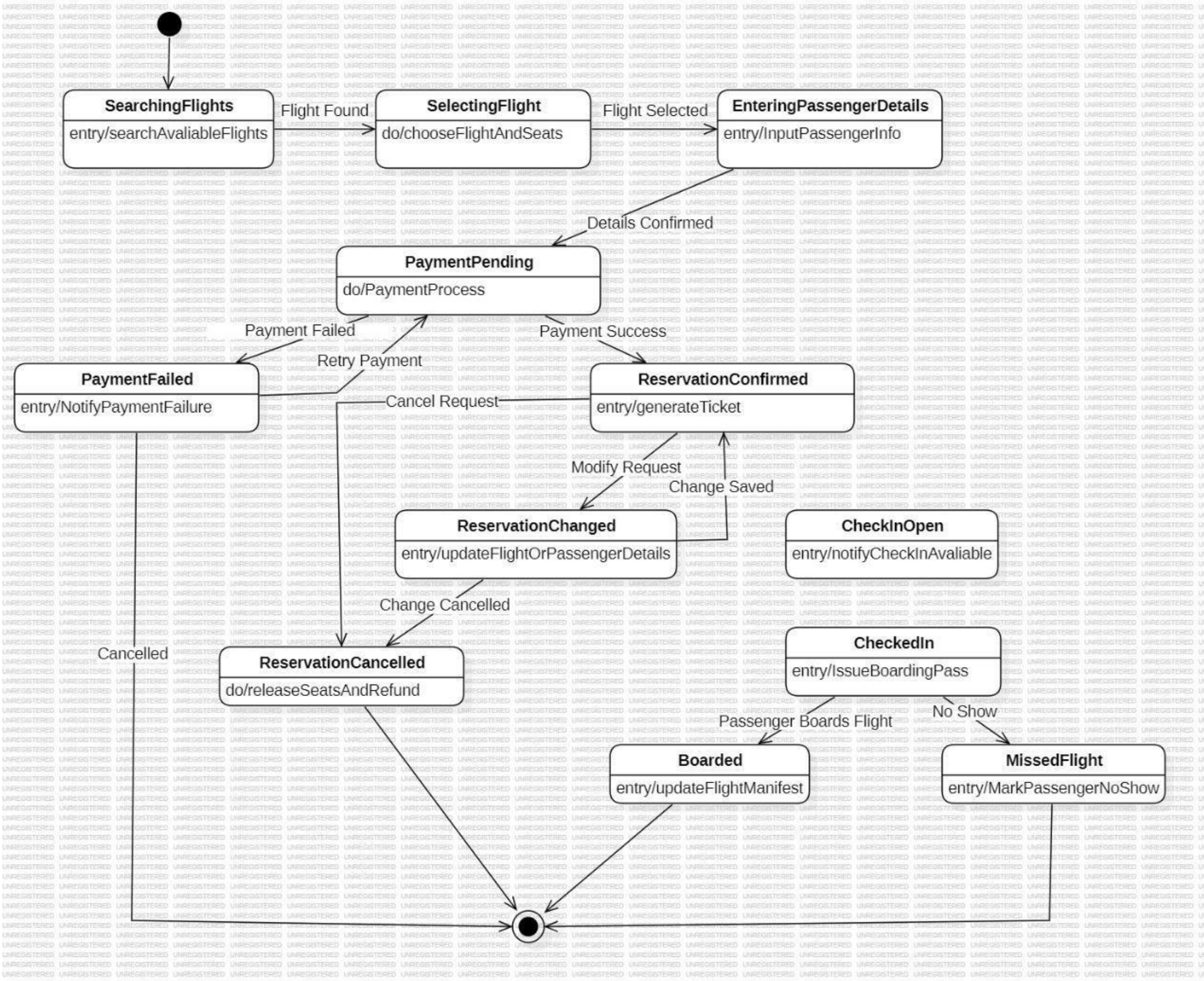
4) A successful payment automatically triggers ticket generation means no manual approval needed.

5) The No Show status is triggered after flight departure time if the passenger has not boarded.

6) A third party payment gateway handles transactions. The system only receives “Payment Success” or “Payment Failed” signals.

7) Payment failures are due to external factors (bank rejection, timeout) and not internal logic errors.

Diagram:



SearchingFlights
entry/searchAvaliableFlights
Flight Found
SelectingFlight
do/chooseFlightAndSeats
Flight Selected
EnteringPassengerDetails
entry/InputPassengerInfo
Details Confirmed
PaymentPending
do/PaymentProcess
Payment Failed
Payment Success
Retry Payment
PaymentFailed
entry/NotifyPaymentFailure
Cancel Request
ReservationConfirmed
entry/generateTicket
Modify Request
Change Saved
ReservationChanged
entry/updateFlightOrPassengerDetails
CheckInOpen
entry/notifyCheckInAvaliable
Change Cancelled
CheckedIn
entry/IssueBoardingPass
Cancelled
ReservationCancelled
do/releaseSeatsAndRefund
Passenger Boards Flight
No Show
Boarded
entry/updateFlightManifest
MissedFlight
entry/MarkPassengerNoShow

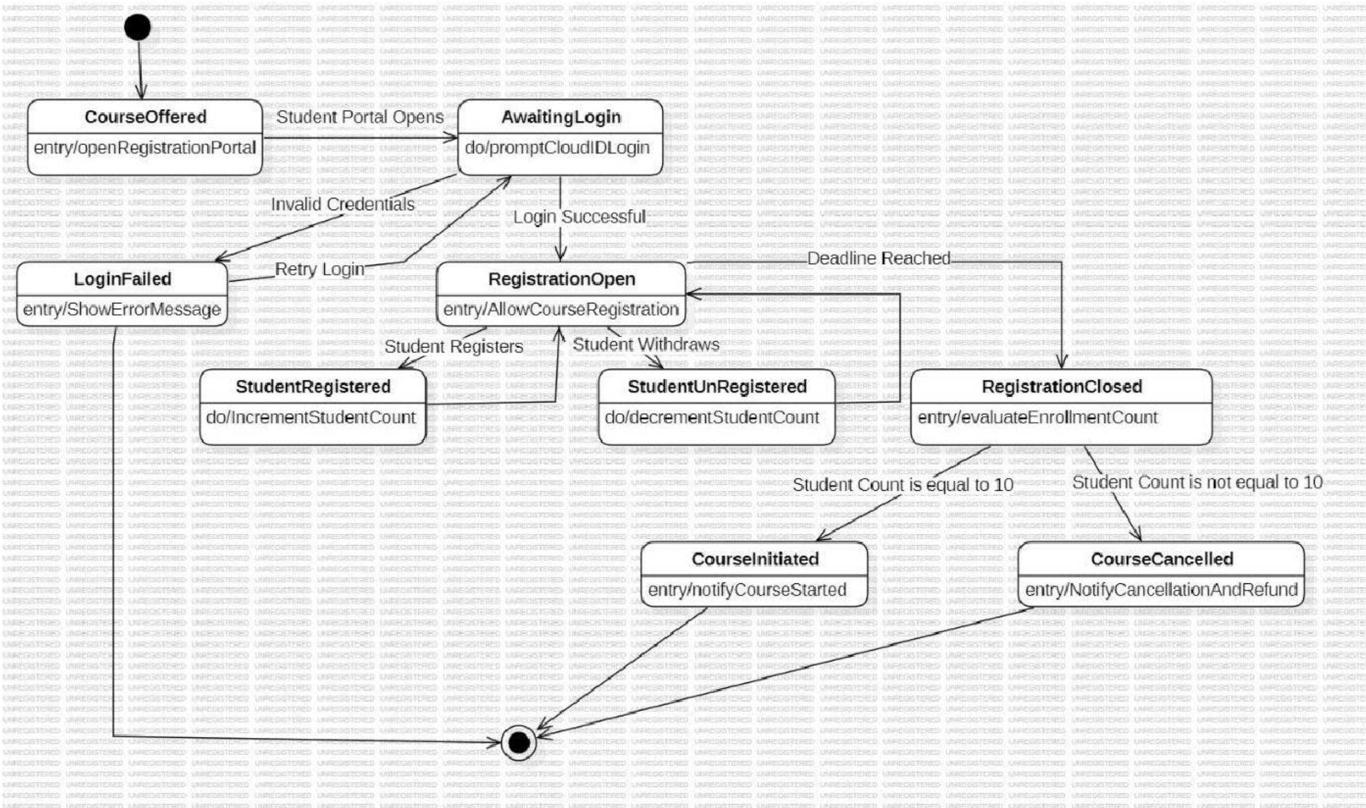
2. Draw State Diagram of University for course registration system

Details: NED University offers new course ,course can initiate only if 10 students (exactly) registered in course ,Registration will cancel otherwise, students must have to login with their cloudIDs for registration.

Asssumptions:

- Course registration opens only when a course is completely published by an academic department, meaning that instructor assignment, schedule, and course capacity have already been determined prior to the time when students can register for that course.
- Students log in to the registration system with their institutional credentials, and access is granted only upon successful verification through the central authentication system of the university.
- If a student enters wrong credentials, the system immediately denies access, showing an error message and allowing the student to retry as long as the registration period is open.
- The registration process remains open for a fixed time window defined by the academic administration; after this window, the system automatically stops any new registration or withdrawal.
- This implies that during the active registration period, a student can only register for the course once, and every successful registration increases the total number of enrolled students instantaneously.
- Only a registered student can withdraw from the course before the registration deadline, and every withdrawal instantly reduces the total count of enrolled students.
- At the close of the registration period, the system verifies the number of registered students and proceeds based on whether that count meets the class size criteria or not.
- The course proceeds only if exactly ten students are registered; any number other than ten leads to cancellation, so that the institution’s fixed-size policy for efficient resource use is reflected.
- If the required number of students is attained, then the class is confirmed; notifications are sent to the students and instructors alike that the course will proceed as planned.
- If the number of students required by the course is not reached, it is cancelled automatically; registered students are informed about cancellation and options of refund or re-enrolment.
- The system is designed to accurately keep a record of every registration, every withdrawal, and every state transition-with no data corruption or duplication, even when hundreds of students interact with the system daily.
- Any network failure or unexpected interruption to the system automatically restores the last saved state and redirects the user back to the login page without losing previous progress.
- Multiple students can attempt to register simultaneously. The system manages all the updates one by one to avoid any wrong or partial overlapping changes in the total enrollment count.
- University administrators can trace the registration process, but nothing allows the manual correction of student headcounts or a bypass of the system's automated decision-making unless officially authorized to reopen. 15. Process Completion The entire registration cycle is considered complete, and once the course either starts or is cancelled, no further modifications, registration, or withdrawal are allowed afterward.

Diagram:



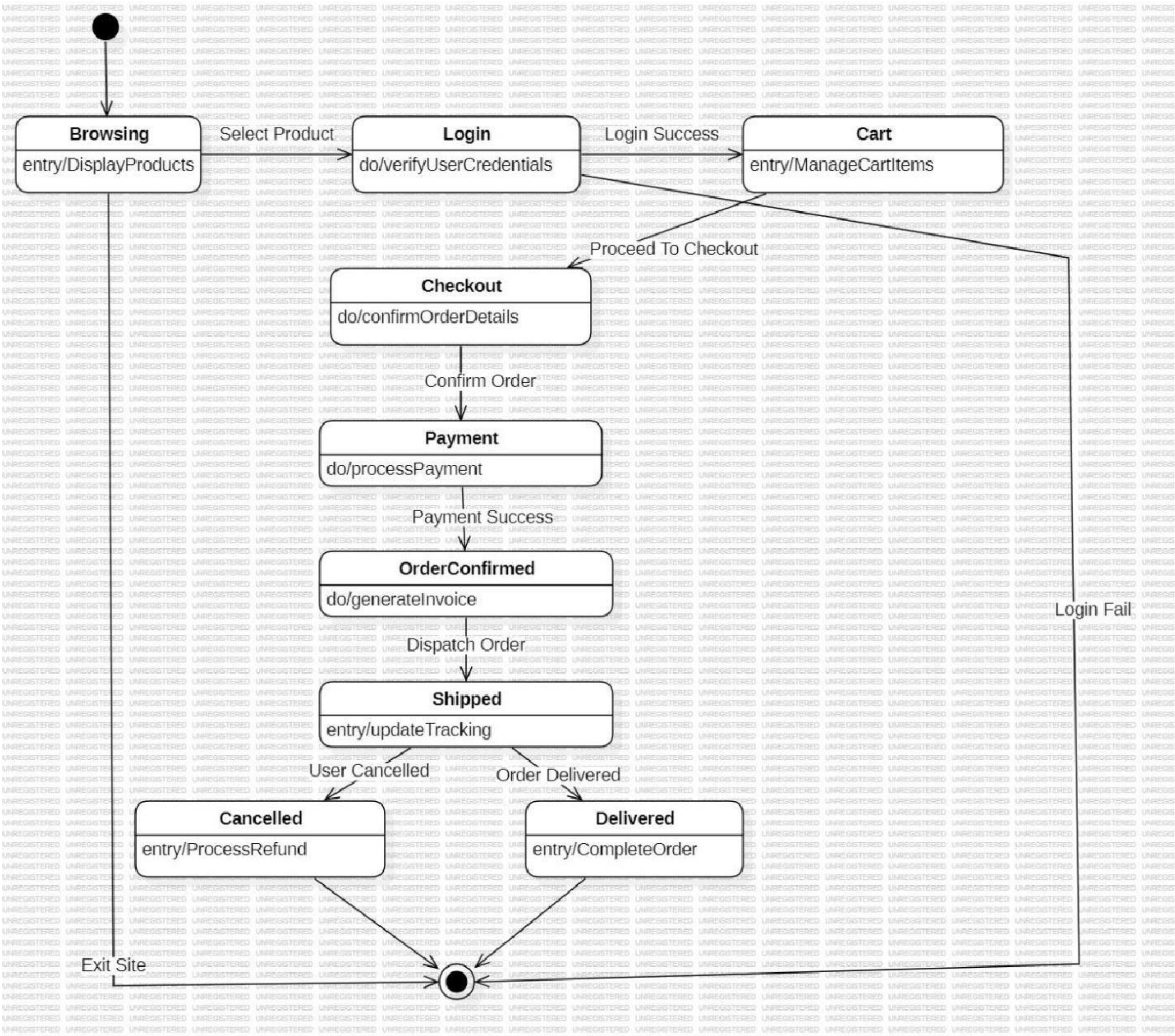
CourseOffered
entry/openRegistrationPortal
Student Portal Opens
AwaitingLogin
do/promptCloudIDLogin
Invalid Credentials
Login Successful
LoginFailed
entry/ShowErrorMessage
Retry Login
RegistrationOpen
entry/AllowCourseRegistration
Deadline Reached
Student Registers
Student Withdraws
StudentRegistered
do/IncrementStudentCount
StudentUnRegistered
do/decrementStudentCount
RegistrationClosed
entry/evaluateEnrollmentCount
Student Count is equal to 10
Student Count is not equal to 10
CourseInitiated
entry/notifyCourseStarted
CourseCancelled
entry/NotifyCancellationAndRefund

3. Draw State Diagram of online shopping system. Consider any website and define states and their transition to successfully place order, also define appropriate start and exit states. Also clearly write down your assumptions.

Assumptions:

1. The user can interact with the E-commerce system through online website or mobile app interface.
2. All product details such as price, description etc and stock information are pre-stored and regularly updated in the product database.
3. Login authentication is managed by secure external authentication service.
4. The user must be successfully logged in before accessing the cart or proceeding to checkout.
5. Payments are processed securely using an integrated third-party payment gateway.
6. An order is confirmed only after the payment transaction is successfully completed.
7. The system automatically generates an invoice once the order is confirmed.
8. Shipping details and tracking updates are received from an external courier service.
9. Users are allowed to cancel their orders only before the product is delivered.
10. Refunds for cancelled orders are processed automatically by the system.
11. Notifications such as order confirmation, shipping updates and delivery status are sent automatically to the user.
12. Any authentication errors, or payment related issues are outside the scope of this state diagram.

Diagram:


Browsing
entry/DisplayProducts
Select Product
Login
do/verifyUserCredentials
Login Success
Cart
entry/ManageCartItems
Proceed To Checkout
Checkout
do/confirmOrderDetails
Confirm Order
Payment
do/processPayment
Payment Success
OrderConfirmed
do/generateInvoice
Dispatch Order
Shipped
entry/updateTracking
User Cancelled
Order Delivered
Cancelled
entry/ProcessRefund
Delivered
entry/CompleteOrder
Login Fail
Exit Site

SDA LAB 6

1. Draw Activity Diagram of the following:

Suppose user can search books and view books online without paying any fee, but he can download book

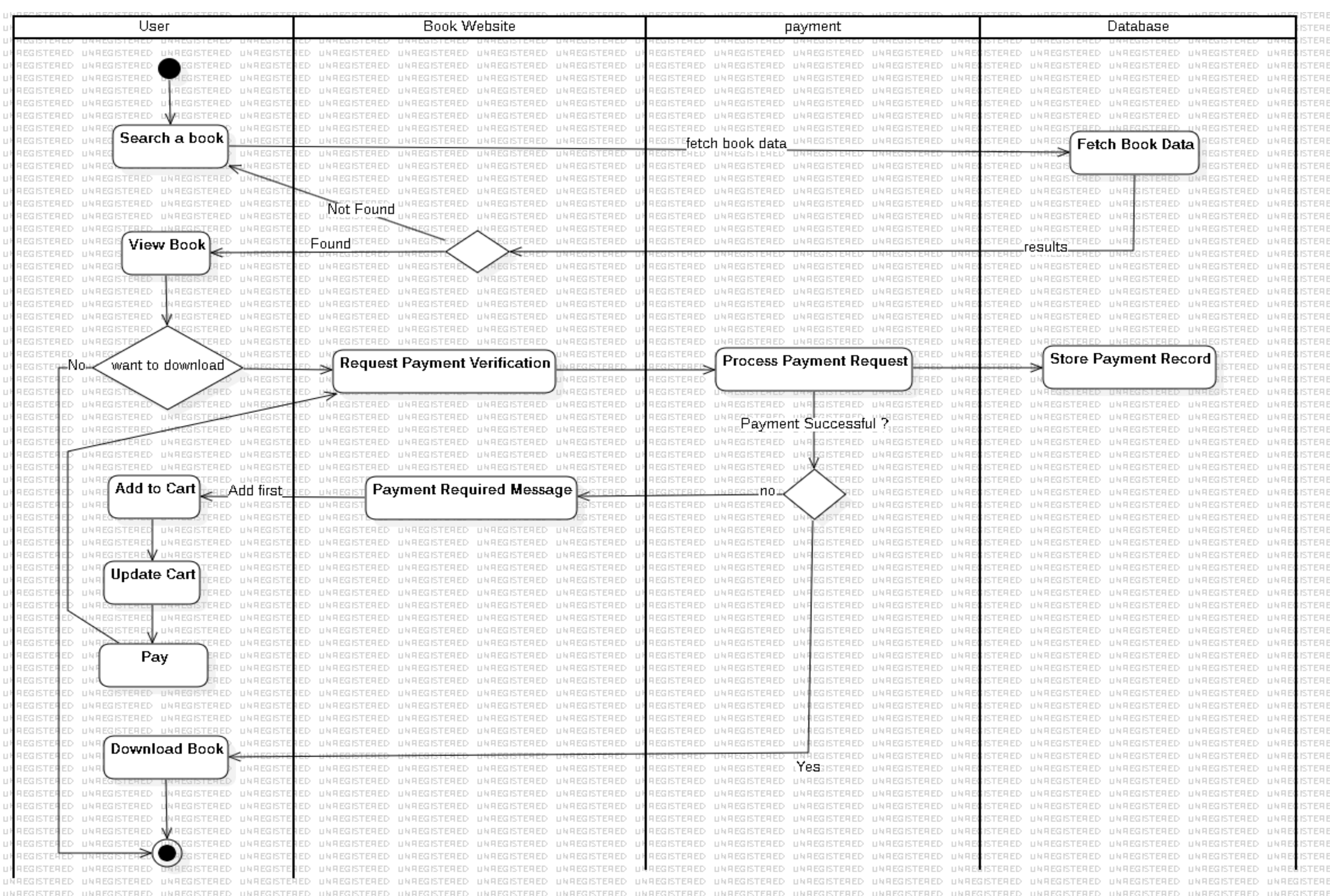
only if he pays , provided the facility to managing cart by using update feature. Make your own assumptions

to complete the scenario and attach printout here

Assumptions:

1. The user searches for a book, and the website fetches book data from the database.
2. If the book is found, the user can view it online or choose to download it.
3. When the user wants to download, the system verifies payment status through the payment service.
4. If payment is required, the user adds the book to the cart, updates it, and completes payment.
5. After successful payment, the book is available for download, and payment details are stored in the database.

Diagram:



2. Draw Activity Diagram of scholarship approval system , make all necessary assumptions and attach printout here

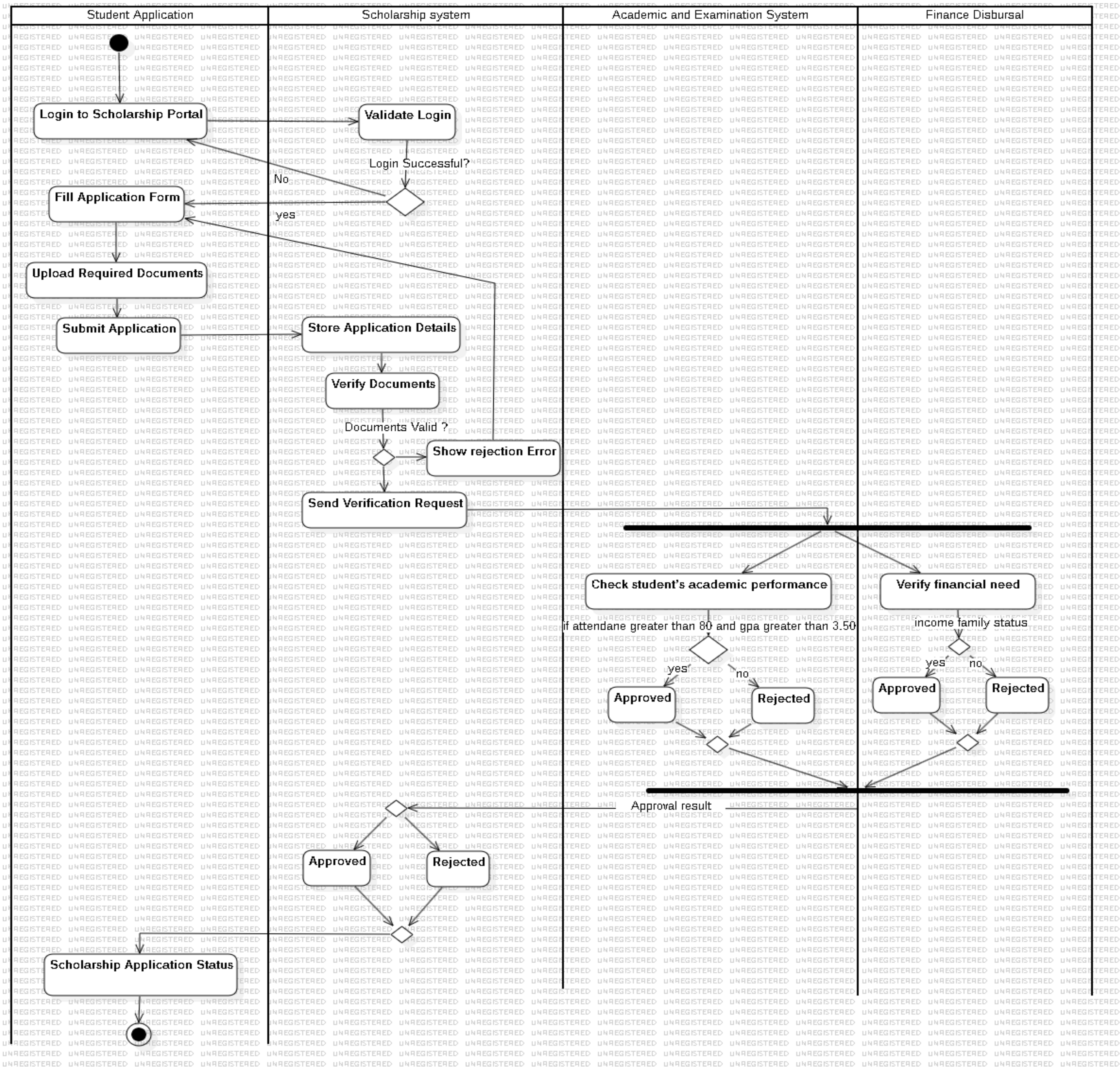
<u>Assumptions:</u>

1. The student fills in the application form with personal and academic details.
2. The student uploads required documents .
3. After submission, the Scholarship System receives the application.
4. The system verifies document completeness and validity (e.g., CNIC format, transcript authenticity).
5. If any document is missing or invalid, the system displays a “Document Error” message.
6. The student is then redirected to fill and resubmit the application form.
7. If the application is complete and valid, the system forwards it for further evaluation.
8. The Academic & Examination Department checks the academic performance of the student.
9. The academic criteria require: Attendance ≥ 80% and CGPA ≥ 3.5.
10. If the student fails to meet academic criteria, the application is rejected and the student is notified.
11. If the student passes academic criteria, the system sends the data to the Finance Department.
12. The Finance Department verifies the student’s financial background.
13. Financial criteria include checking:
14. Family income status:

- Parent/guardian’s income slips
- If the student does not meet the financial need requirement, the scholarship is rejected.
- If both academic and financial criteria are satisfied, the Scholarship System approves the scholarship.

15. The process ends when the scholarship is either approved and disbursed or rejected.

Diagram:


Student Application
Scholarship system
Academic and Examination System
Finance Disbursal
Login to Scholarship Portal
Validate Login
Login Successful?
No
yes
Fill Application Form
Upload Required Documents
Submit Application
Store Application Details
Verify Documents
Documents Valid ?
Show rejection Error
Send Verification Request
Check student's academic performance
Verify financial need
if attendane greater than 80 and gpa greater than 3.50
income family status
yes
no
Approved
Rejected
yes
no
Approved
Rejected
Approval result
Approved
Rejected
Scholarship Application Status

3. Suppose we want to develop a web based social media network with the following functionalities:

- **The user should sign up for the system**
- **The user should log into the system and can send friend request or block the user**

Write your assumptions against each activity and Draw activity diagram.

Hint:

Activity1: signup

Activity 2: login

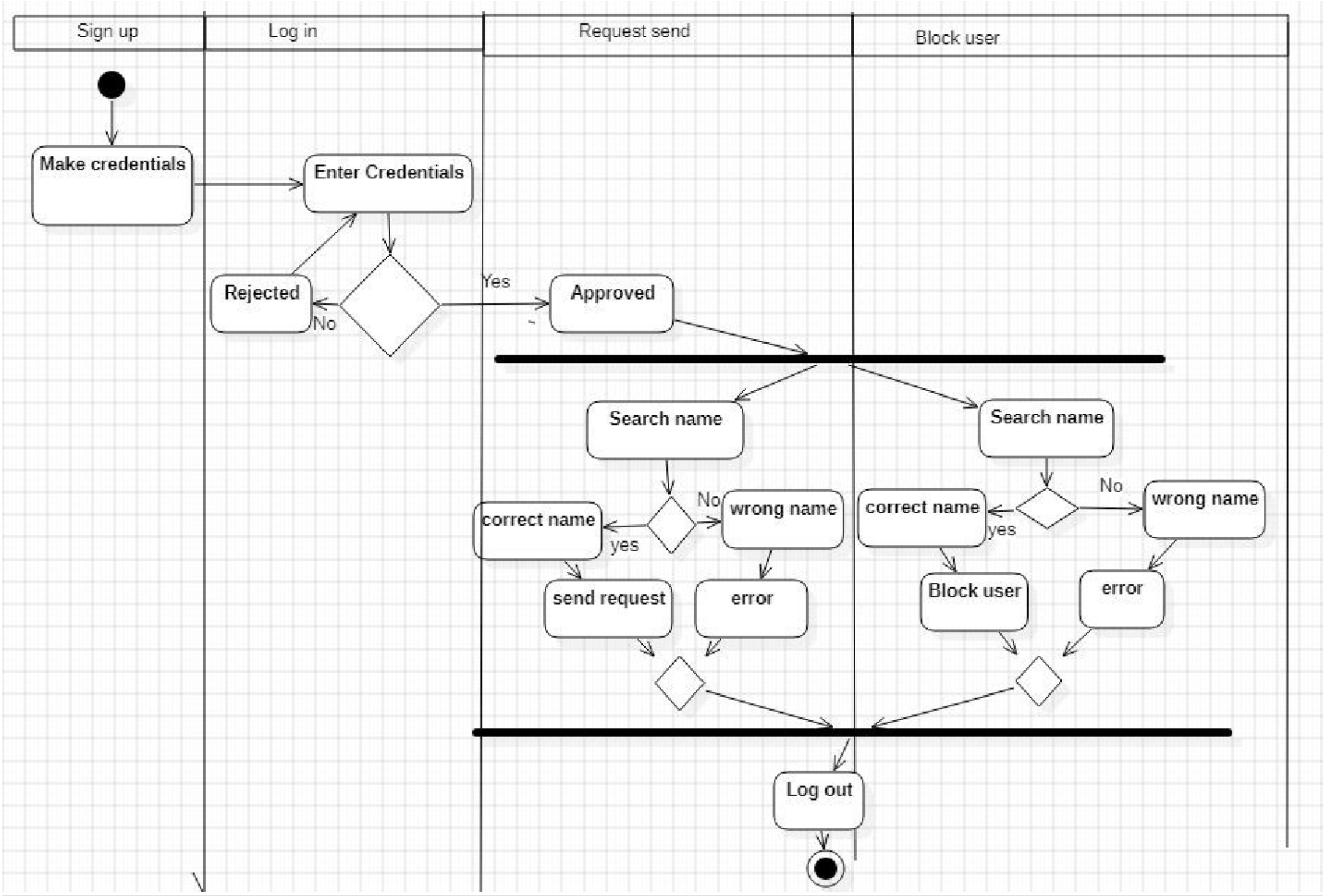
Activity3: send friend request

Activity4 :block user

<u>Assumptions:</u>

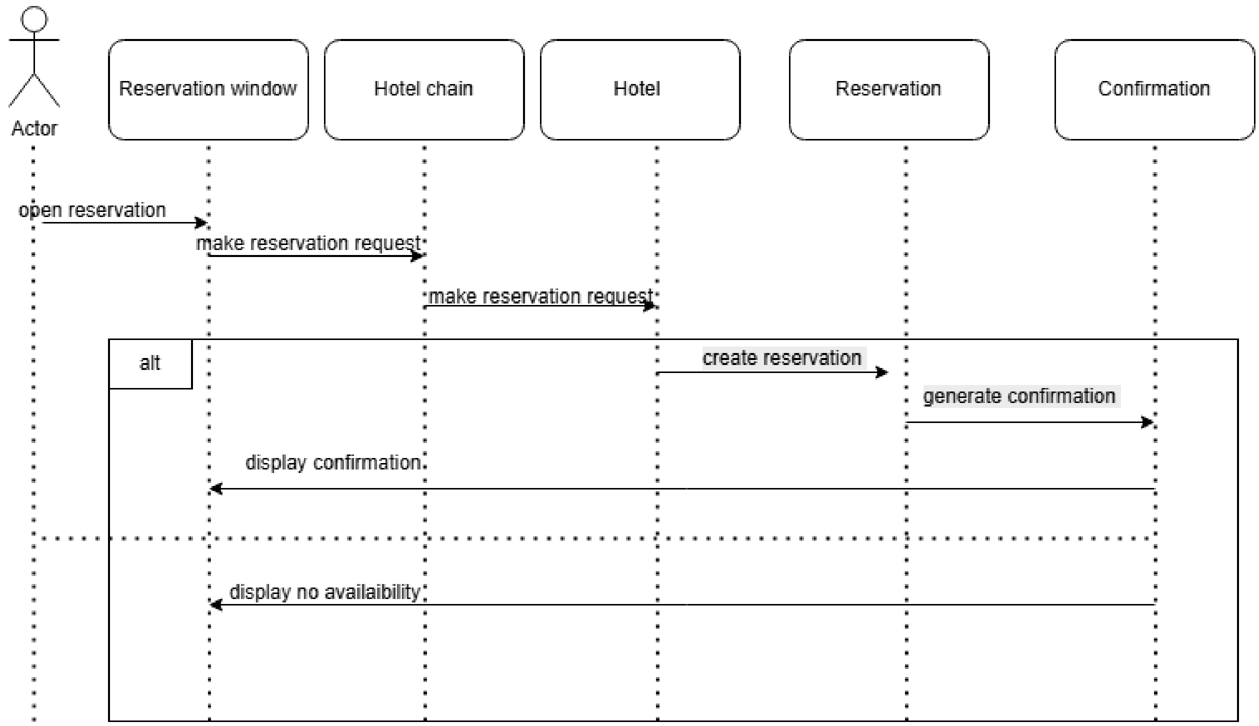
1. A central user database exists for credential checks. It enables searching and validating other users by name.
2. The user must be logged in to block someone. The "Block user" flow starts post-authentication.
3. All process steps are sequential and dependent. One action must finish before the next can begin.
4. The system distinguishes a "wrong name" from other errors. It provides specific feedback for a failed user search.
5. "Log out" is a common endpoint for many process paths.

Diagram:


Sign up
Log in
Request send
Block user
Make credentials
Enter Credentials
Rejected
No
Yes
Approved
Search name
Search name
correct name
yes
No
wrong name
correct name
yes
No
wrong name
send request
error
Block user
error
Log out

Lab Session 07

1. Draw a sequence diagram for Hotel Reservation System Hint: The object initiating the sequence of messages is a Reservation window. The Reservationwindow sends a makeReservation() message to a HotelChain. The HotelChain then sends a makeReservation() message to a Hotel. If the Hotel has available rooms, then it makes a Reservation and a Confirmation. Explore the scenario and Write Down your assumptions also attach printout of diagram here.



<u>Assumptions:</u>

1. Initiating a Reservation Request

The request-generation process begins when a customer or user initiates a reservation request through the online reservation window, assuming the interface is already available and accessible to all users.

2. User Authentication

It is expected that entry to the reservation window comes from a user already logged in or authenticated; it ensures that only a verified user can create booking requests.

3. Communication with Hotel Chain System

The moment the user clicks to open the option for reservation, the reservation window automatically forwards the request to the central hotel chain system that deals with all its hotels.

4. Forward Request to the Hotel in Question

The hotel chain system sends the booking request it receives to a particular hotel based on the user's selected location or preferences or availability data stored with the system.

5. Hotel System Availability Check

The hotel system checks the availability of rooms according to the required dates, room type, and occupancy details provided by the user before proceeding with the reservation.

6. How the Reservation Creation Process Works

If the rooms are available, the hotel system will create a reservation entry that contains customer details, dates of a booking, and information about a payment or hold.

7. Confirmation Generation

Here, once the reservation has been created, the booking confirmation will be generated by the hotel system, which includes a reservation ID, check-in/check-out dates, and hotel contact information.

8. Displaying Confirmation to User

Once confirmation is generated, it is sent back through the hotel chain system to the reservation window and displayed as proof of a successful booking in front of the user.

9. Handling No Availability Scenario

In case the hotel system does not have the requested room type or date range, it returns a "no availability" response without creating the reservation, which is shown in the reservation window to the user.

10. Alternative Flow Execution

The “alt” block within this diagram shows only one of the two responses—either confirmation or no availability is executed depending on the result of the availability check.

11. Data Synchronization Between Systems

It is assumed that different involved systems, such as the reservation window, chain, and individual hotel, are synchronized in real time regarding the availability of hotel rooms and the exact status of the booking.

12. Reliability of Communication Channels

It assumes that communications between the reservation window, hotel chain, and hotel systems never introduce any delay, failure, or loss of messages, so that transactions can smoothly flow.

13. Reservation Storage and Record Keeping

Every confirmed reservation is stored securely within the hotel's central reservation database for keeping, auditing, and future customer reference.

14. Timing of User Notification

The confirmation or unavailability message should appear to the user immediately after the hotel system completes the reservation or determines the unavailability to provide real-time feedback.

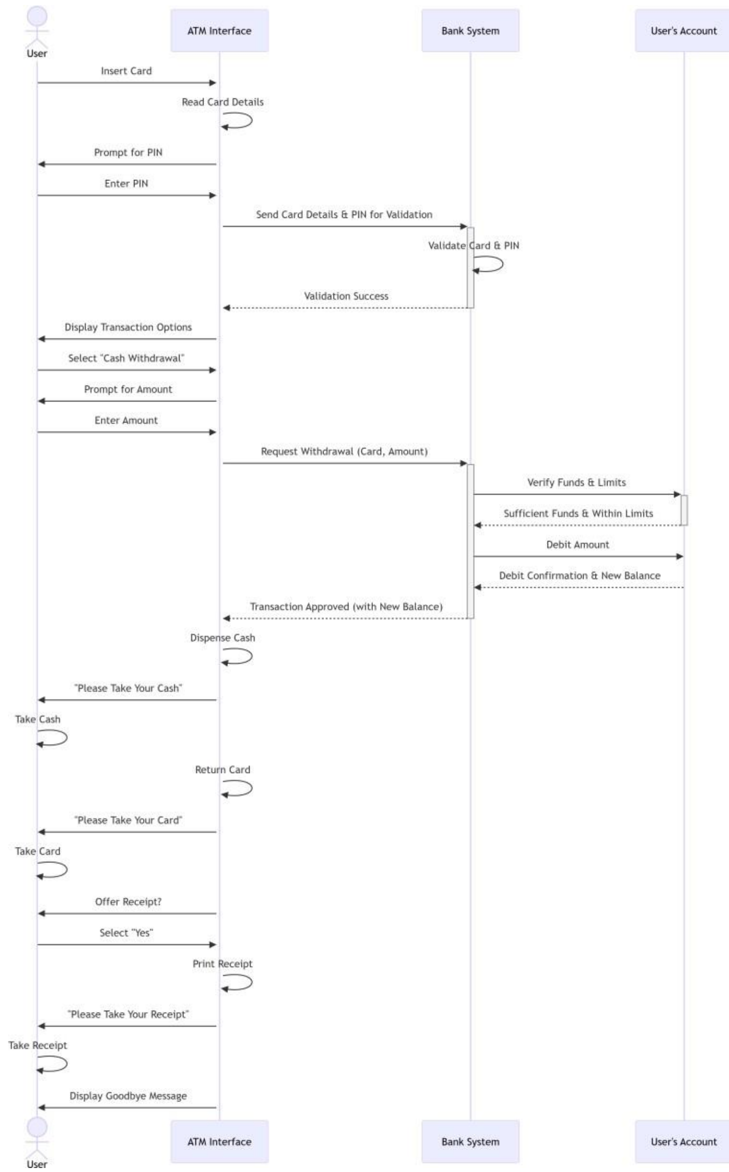
15. Administrative Oversight

It is assumed that hotel administrators are able to view, update, or cancel reservations through a separate management interface, but such manual operations are outside the scope of this diagram.

16. End The reservation process will be concluded once the customer receives either the booking

confirmation or the unavailability message, which also ends the interaction within this sequence.

2. Consider Cash withdrawal through ATM and construct sequence diagram . Explore the scenario and Write Down your assumptions also attach printout of diagram here



Assumptions:

1. Card Insertion

The withdrawal process initiates with the insertion of a valid ATM or debit card into the machine, considering that the card reader operates and can read data from either chip or magnetic stripe of the card.

2. User Authentication

The system assumes that the customer should enter a correct PIN in order to proceed, and this information is securely sent from the ATM to the bank server.

3. PIN Verification

Any entered PINs are checked against encrypted records in the bank's account database; therefore, no local verification has taken place at the ATM to maintain security integrity.

4. Transaction Authorization

The system allows the customer to select a transaction type, like cash withdrawal, balance inquiry, or transfer, after successfully authenticating him/her.

5. Withdrawal Application

Assuming the amount entered is a valid multiple of available denominations, the customer selects the “Cash Withdrawal” option and enters the desired amount.

6. Communication with Bank Server

Assuming there is a stable and encrypted connection to prevent the interception of data, the ATM forwards the request for withdrawal via a secure network connection to the bank server.

7. Account Verification

The bank server checks the customer's account balance in the central database to see if enough funds are available for approval of the transaction.

8. Insufficient Balance Handling

If the requested amount exceeds the account's balance, the bank server will reject the transaction with an error message that is issued to the customer via the ATM.

9. Cash Dispensing

If the request for withdrawal is granted, the ATM machine should then tell the cash dispenser module to dispense, accurately, the requested amount of money to the customer.

10. Recording of Transactions

At the same time, the bank server updates the balance of the customer’s account and records all transaction details in the transaction log for future reference and auditing.

11. Generating Receipts

Assuming the receipt printer has paper, a receipt would be printed by the ATM, showing details like the date and time of the transaction, the amount withdrawn, and the remaining balance.

12. Card Ejection

Once the transaction is completed, the ATM ejects the card and asks the customer to take it back for the session to end securely.

13. Session Timeout

The ATM cancels the transaction and ejects the card automatically as a security measure if no response is given by the customer after a certain time period.

14. Error Handling

Upon failure of communication with the bank's server, or in case of any internal error, the ATM would flash an appropriate error message and end the session gracefully.

15. Security and Privacy

It is assumed that all communication between the ATM, the bank server, and the account database is encrypted, and that no sensitive data, including PINs, are stored locally on the ATM.

16. Termination of Transaction The process is complete when the customer gets cash and the system successfully records the transaction in both the ATM and bank databases.

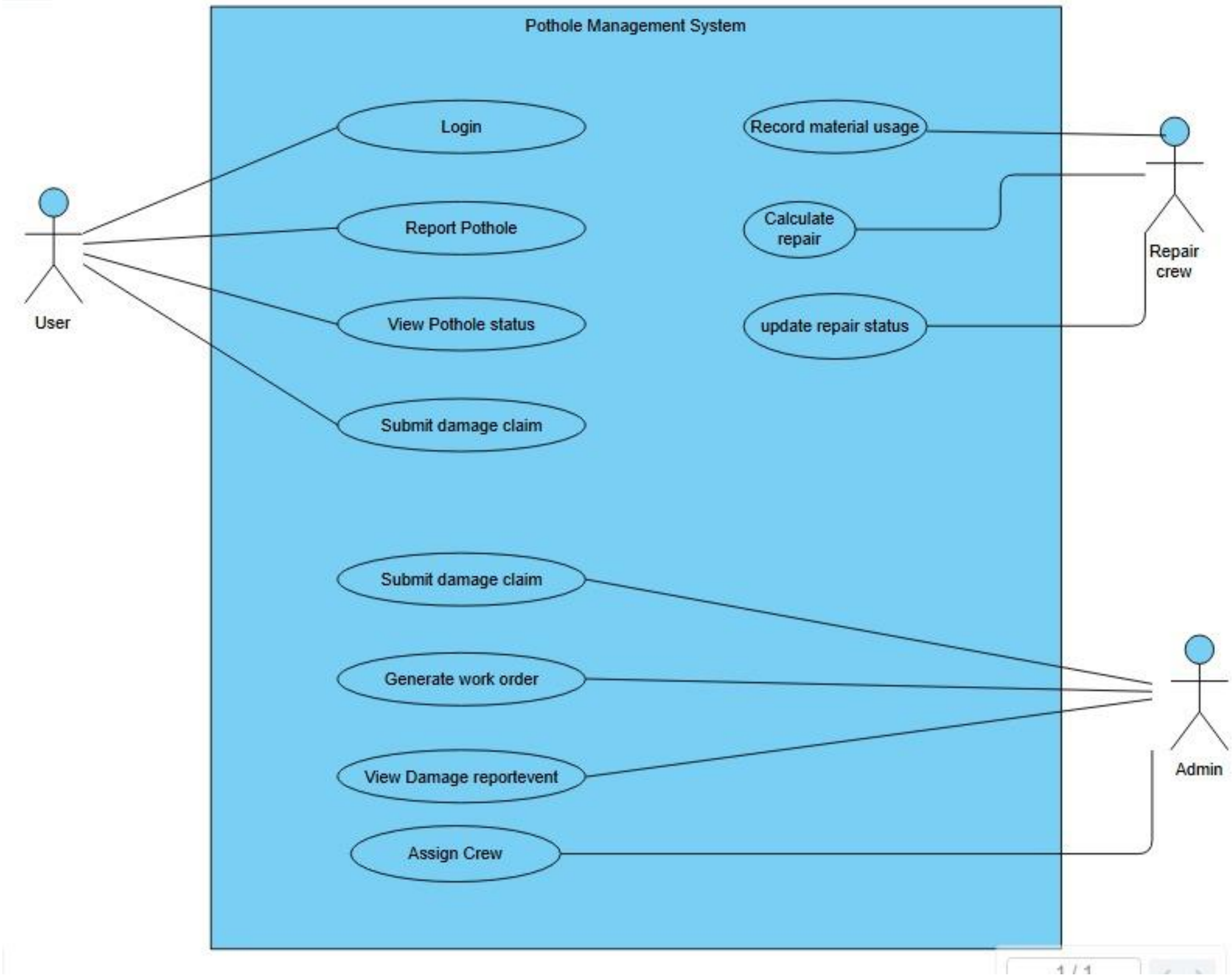
Lab Session 08

1. Write down the number of assumptions about the manner in which user interact with the system

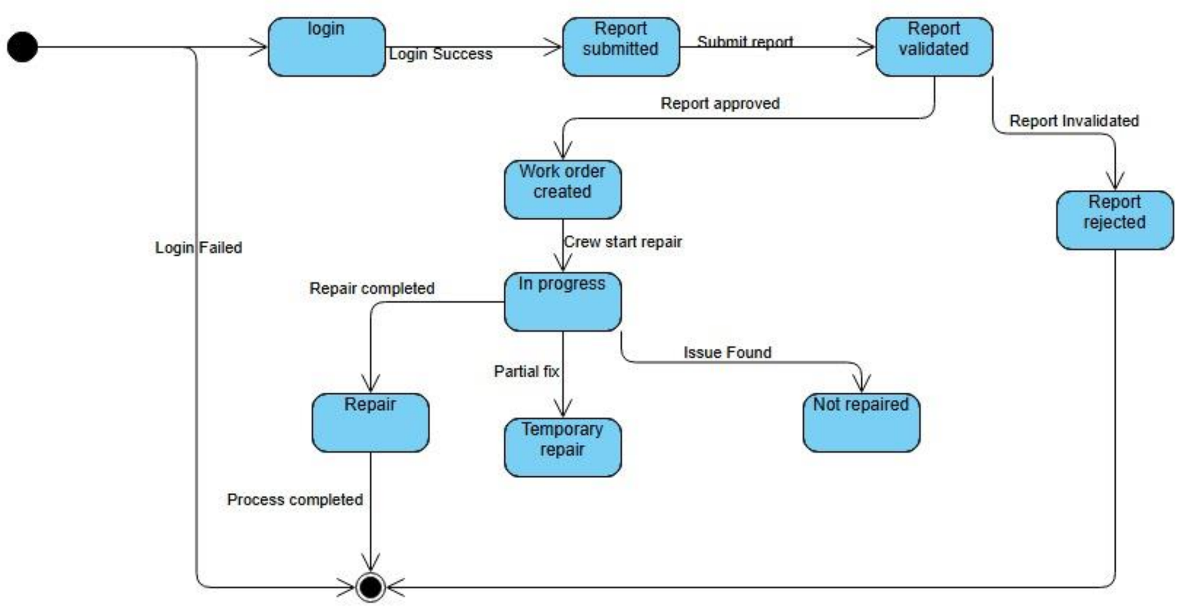
Ans:

- User must login for accessing the system
- Each pothole is given unique id automatically
- Repair priority is computed from pothole severity
- Citizen can view and tracks only their own reports
- All action occur through a web base interface
- Officer manually assign crew and Equipements
- Crew can updated status
- Admin mange user and oversees all data

Q2:Draw a UML USE case diagram for this system



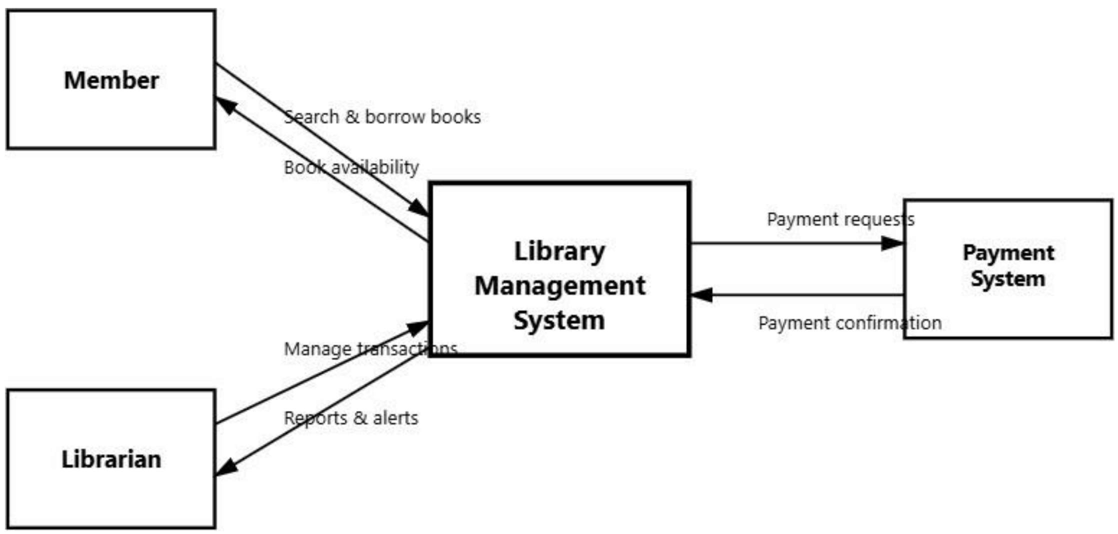
Q3:Draw A State diagram for this system



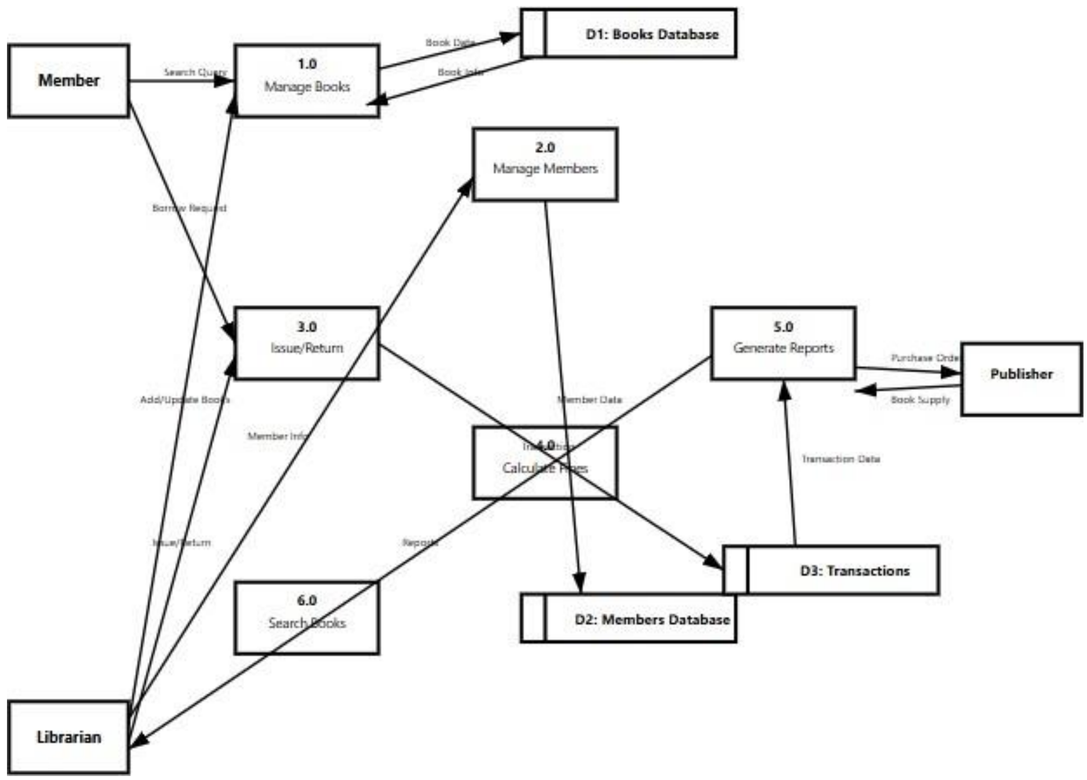
Lab Session 09

2. Draw context diagram and Level 1 diagram of Library Management system, write down requirements and (Attach printout here).

CONTEXT LEVEL:



LEVEL 1:



REQUIREMENTS:

Functional Requirements

1. The system shall allow librarian to add new books with ISBN, title, author, publisher, edition, and category details.

2. The system shall allow librarian to update existing book information in the database.
3. The system shall allow librarian to delete books from the inventory.
4. The system shall track book availability status as available, issued, or reserved.
5. The system shall maintain multiple copies of the same book with unique identifiers.
6. The system shall allow librarian to register new members with personal details including name, address, contact number, and email.
7. The system shall issue unique library card number or member ID to each registered member.
8. The system shall allow librarian to update member information.
9. The system shall track complete borrowing history for each member.

10. The system shall manage different member categories such as student, faculty, and public with different borrowing privileges.
11. The system shall allow librarian to issue books to registered members.
12. The system shall set due dates automatically based on member category.
13. The system shall record return date and book condition when books are returned.
14. The system shall allow members to renew borrowed books if no reservations exist.
15. The system shall allow members to reserve books that are currently issued.
16. The system shall automatically calculate fines for overdue books based on number of days delayed.
17. The system shall apply different fine rates for different member categories.
18. The system shall record fine payment transactions with date and amount.
19. The system shall generate fine receipts for members.
20. The system shall allow users to search books by title, author, ISBN, or category.
21. The system shall display book availability status in real-time during search.
22. The system shall display complete book details including location in library.
23. The system shall provide advanced search with multiple filter criteria.
24. The system shall generate reports of overdue books with member details.
25. The system shall generate reports of most issued books in specified time period.
26. The system shall generate member activity reports showing borrowing patterns.
27. The system shall generate daily, weekly, and monthly transaction reports.
28. The system shall generate fine collection reports with total amounts.
29. The system shall allow librarian to create purchase orders for publishers.
30. The system shall track book supplies received from publishers

Non-Functional Requirements

1. System should respond to search queries within 2 seconds and handle at least 100 concurrent users
2. Role-based access control, encrypted passwords, secure login authentication, and regular data backups
3. System uptime of 99.5% with automatic error logging and recovery mechanisms

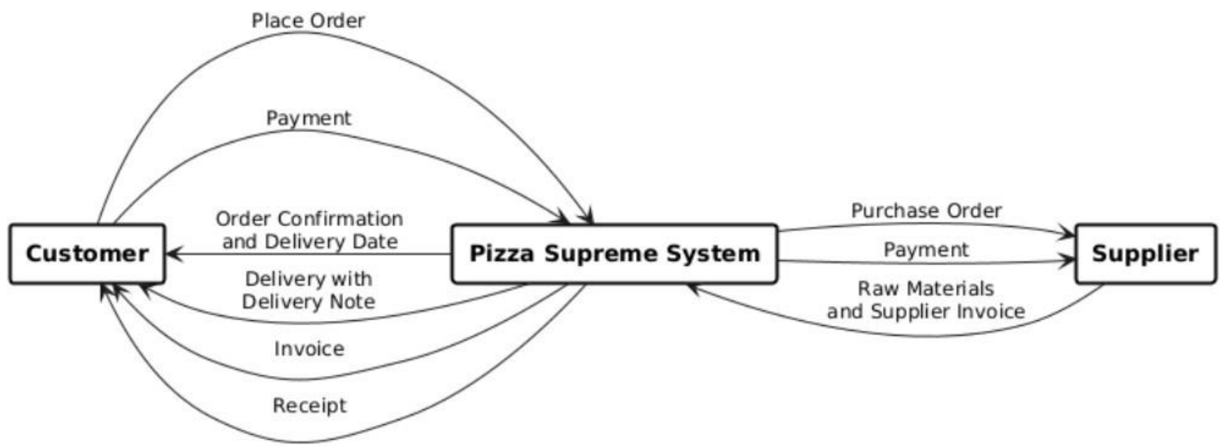
Intuitive user interface, minimal training required, accessible from multiple devices

4. Capable of handling growth in books, members, and transactions without performance degradation
5. Modular design for easy updates and bug fixes with comprehensive documentation

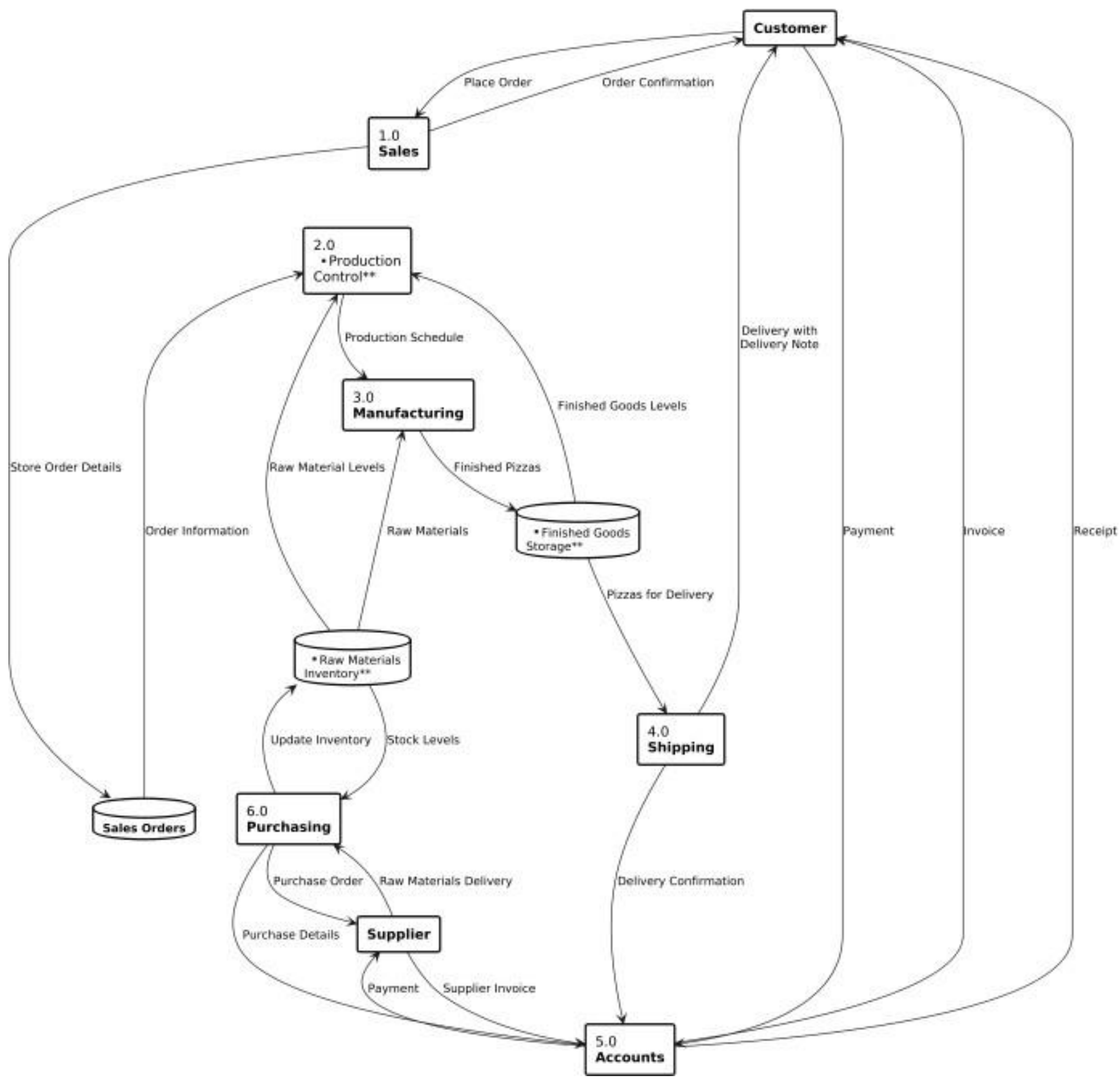
3. Draw context diagram and Level 1 diagram of the following case study: Case Study – Pizza Supreme A large pizza business makes pizzas and sells them. The pizzas are manufactured and kept in cold storage for not more than two weeks. The business is split into a number of functional units. There is Production Control, Manufacturing, Stores, Accounts, Sales, Shipping

and Purchasing. Production Control is responsible for organizing which pizzas to produce in what order and in what quantity. They need to schedule the production of the pizzas according to the current and expected sales orders together with the number of pizzas already in Stores. Manufacturing take the raw materials from the Stores and manufacture pizzas returning the completed goods to the Stores. Accounts deal with the payments for the pizzas when delivered to the customer and the payment to the suppliers of the raw materials. Sales deal with customer orders whilst Purchasing organize the buying of raw material from suppliers. Shipping manages the packing and delivery of the goods to the customer with a delivery note. When a sales order is received by sales they record what is being ordered and by whom. They also record the details of the expected date of delivery. Production Control access this information and make sure that, if required, pizzas are produced by Manufacturing and are ready in Stores for when the delivery needs to be made. After the delivery is made Accounts make sure that the customer receives an invoice and that payment for the invoice is received at which time a receipt is issued. Purchasing look at the current stock of raw materials and by using current stock levels, supplier turnaround times and quantity to be ordered decide what needs to be ordered on a daily basis. Their aim is never to run out of an ingredient but to minimize the amount of raw material kept in stock.(Attach Printout here).

CONTEXT LEVEL:



LEVEL 1:


Customer
Place Order
Order Confirmation
1.0
Sales
2.0
• Production
Control**
Production Schedule
3.0
Manufacturing
Delivery with
Delivery Note
Finished Goods Levels
Store Order Details
Raw Material Levels
Finished Pizzas
Order Information
Raw Materials
• Finished Goods
Storage**
Payment
Invoice
Receipt
Pizzas for Delivery
• Raw Materials
Inventory**
Update Inventory
Stock Levels
4.0
Shipping
Sales Orders
6.0
Purchasing
Purchase Order
Raw Materials Delivery
Delivery Confirmation
Purchase Details
Supplier
Payment
Supplier Invoice
5.0
Accounts

LAB SESSION 10

Q1: Draw deployment diagram of Student Management System, where Student /Staff /and admin can be the external nodes, admin can connect to system to manage and update database activities through private network (say LAN), and staff/students can interact with system through internet. Student database can stores staff data file and student data file. An online database server will provides the customize view of student/staff‛s queries. Mention all required components with their attributes and possible functions inside management system.

Assumptions:

1. Network Connectivity

All student PCs and staff PCs are assumed to have stable Internet access to connect to application servers, and servers are connected to each other over a secure LAN to minimize latency in database operations.

2. Server Role Separation

Each Application Server has a distinct responsibility: one handles update operations and the other takes care of query operations, to reduce load contention and increase performance.

3. Centralized Data Storage

All the student and staff data files are stored on one MySQL Database Server, and all read/write operations from the application servers synchronize to this central instance of the database.

4. Authentication Control

Both students and staff are authenticated through the application servers; this means that user credentials and access levels are regulated by the servers before any data transaction with the database.

5. Handling Concurrent Access

This is where it is assumed that the application servers implement concurrency control to handle simultaneous requests, such as multiple students updating or querying data, without corrupting or duplicating records.

6. Communication protocols

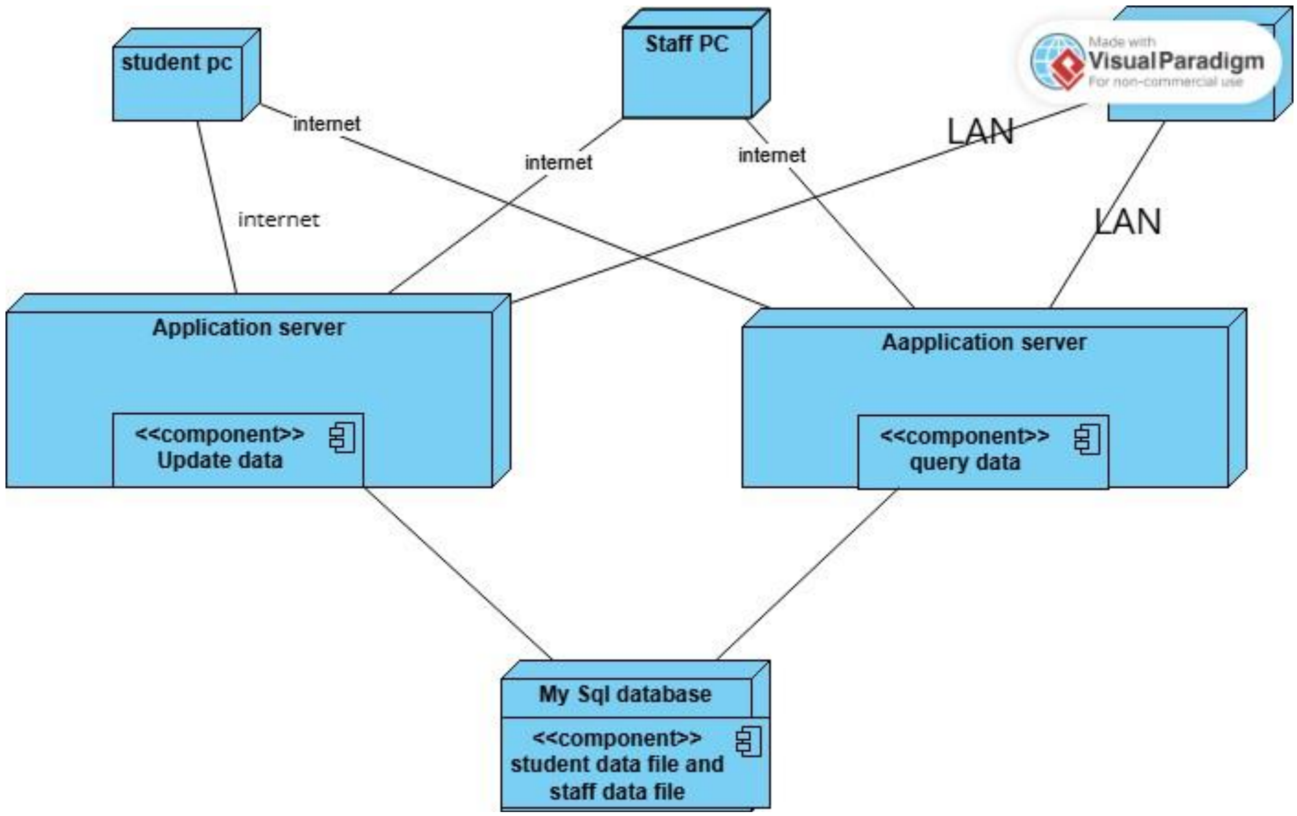
All the communication between clients and servers is done over secure HTTP/HTTPS, while serverto-database communication uses the standard MySQL protocol within the LAN for better speed and security.

7. Data Consistency Policy

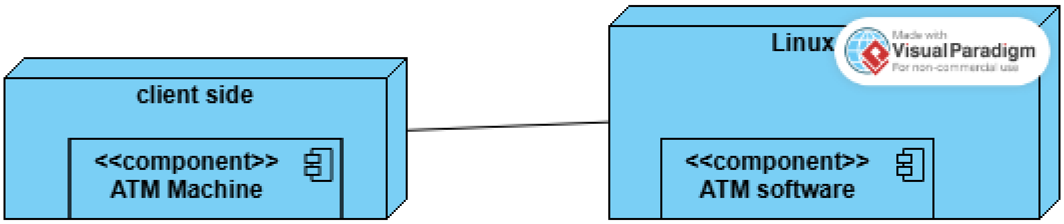
Updates made via the update server will be immediately reflected in the query server's responses, and therefore the update model is real-time as opposed to batch updates.

8. Fault Tolerance

If one of the application servers fails-for example, an update server-it is assumed that the system has built-in failover or load-balancing mechanisms that would still keep the service available.


student pc
Staff PC
Made with
Visual Paradigm
For non-commercial use
internet
internet
internet
internet
LAN
LAN
Application server
<<component>>
Update data
Aapplication server
<<component>>
query data
My Sql database
<<component>>
student data file and
staff data file

Q2: Draw deployment diagram of ATM, ATM software can only execute on Linux operating system, add all possible components of software with attributes and operation, and show interaction with devices involved in the system.



Assumptions:

1. Operating Environment

It is assumed that the server-side ATM software operates on a Linux-based environment, which is chosen for its security, stability, and compatibility with banking middleware and encryption libraries.

2. Network Communication

The communication channel that connects an ATM machine (client side) to the ATM software-which is situated on the server side-must ensure the confidentiality and integrity of user transactions.

3. Role of Client Side

The ATM Machine is merely an input/output interface: it only interacts with the user directly by means of card reading, receipt printing; everything else is handled by the backend software.

4. Server Functionality

Core banking activities, like balance checking and giving withdrawal permission, are all done at the server side by the ATM Software after the request has been validated at the client's end.

5. Real-Time Transaction Processing

It is assumed that all user ATM operations are performed in real-time, with each request instantly updating the corresponding transaction in the back end for the consistency of the account balance.

6. Authentication and Security

User authentication by card and PIN is controlled at the server, and the verification of the PIN is encrypted to avoid exposure on the client side.

7. System Availability

Design assumptions are: the Linux server is highly available and part of a redundant cluster, which ensures continuity even if one server node goes down.

8. Hardware-Software Integration

This means that the ATM Machine hardware-card reader, keypad, and display-communicate via an API for smooth data exchange with the Linux-based ATM software.

LAB 11

Scenario:

1.A customer uses a mobile app to place a food order from a restaurant. The app sends the order details

to the restaurant system, which confirms the order, prepares the food, and notifies the delivery service.

The delivery service assigns a driver, picks up the food, and delivers it to the customer.

Instructions:

a. Identify the objects/instances involved in this process (e.g., Customer, MobileApp,

RestaurantSystem, DeliveryService, Driver).

- Show the sequence of messages exchanged to complete the process:
- Customer selects items and places an order via the Mobile App.
- Mobile App sends the order to the Restaurant System.
- Restaurant System confirms order and updates order status.
- Restaurant System notifies Delivery Service.
- Delivery Service assigns Driver and updates status.
- Driver picks up and delivers food to Customer.

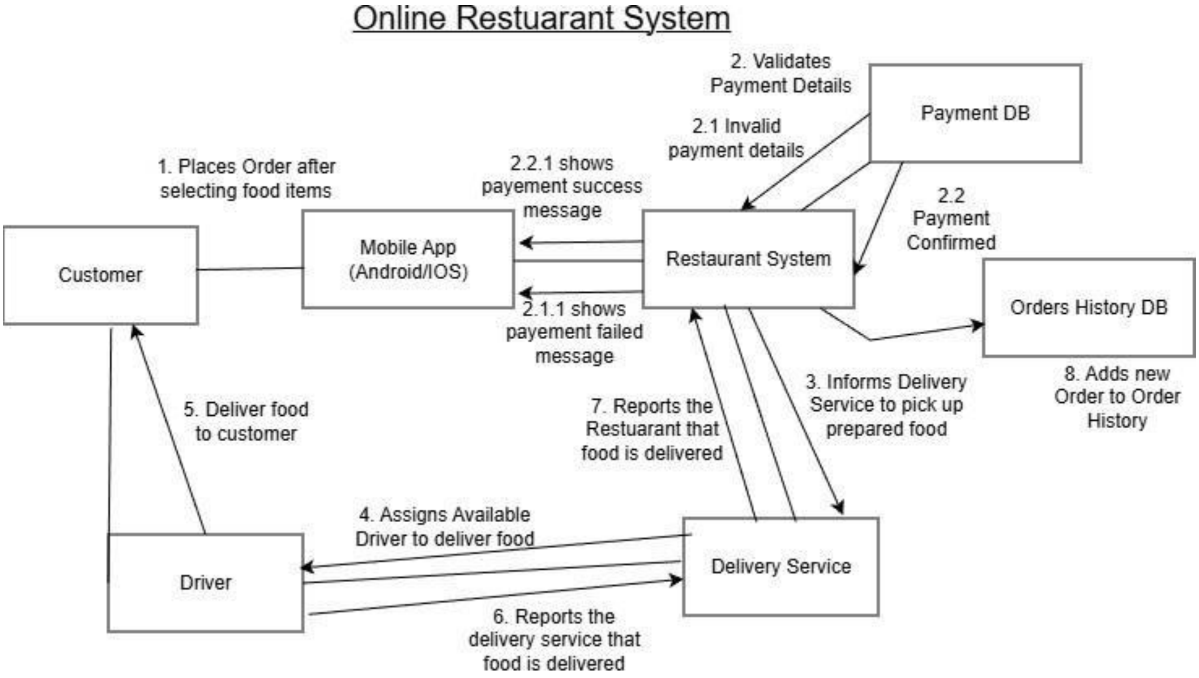
b. Use message numbering (e.g., 1: placeOrder(), 2: sendOrderDetails()) to indicate the order of

interactions.

c. Include return messages where appropriate (e.g., orderConfirmation).

Deliverable:

- A UML Communication Diagram that:
- Shows all involved objects.
- Correctly numbers messages in execution order.
- Clearly illustrates interactions between customer, systems, and services.



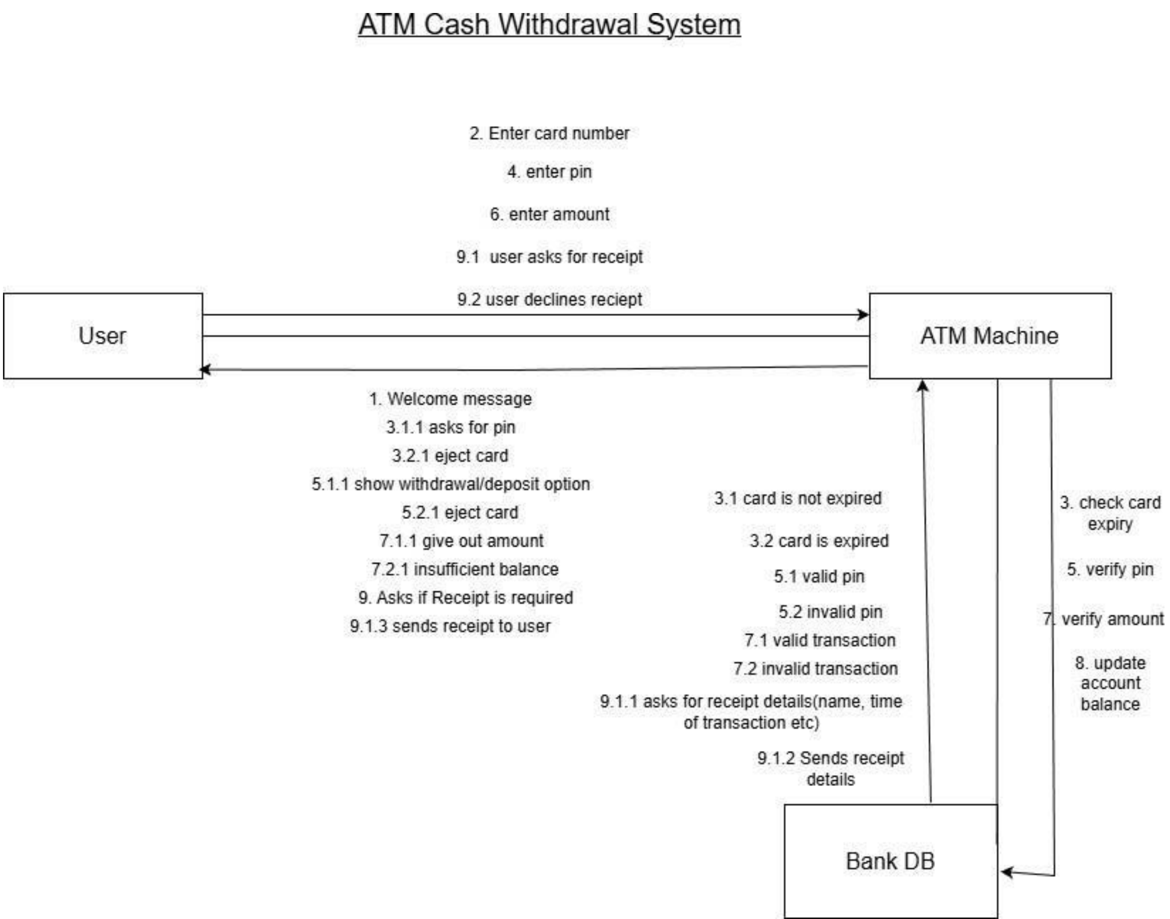
2. Draw communication Diagram of ATM Cash Withdrawal System

Assumptions:

- It initiates the process of withdrawal when a customer inserts a valid ATM or debit card into the machine, provided the card reader of the machine is functioning and capable of reading the chip or magnetic stripe data on the card.
- The system assumes it is necessary for the customer to enter a correct PIN number to continue, where this PIN number is sent securely to the bank server for verification via the ATM.
- The PIN entered is matched for authentication from the bank's account database, which has encrypted records; no local verification of transactions occurs at the ATM in order to maintain security integrity.
- The system prompts the customer to choose a transaction type such as cash withdrawal, balance inquiry, or transfer only after authentication is successful.
- Assume the customer has selected the "Cash Withdrawal" option and entered the desired amount, which shall be a valid multiple of the available denominations supported by the machine.
- Assuming a good network connection, wherein the connection is stable and encrypted to prevent data interception, the ATM will forward the withdrawal request to the bank server.
- The balance of the customer's account is checked by the bank server against the central database for sufficient funds; then it grants the dispensing transaction.

- In case the requested amount exceeds the account's balance, the bank server disallows the transaction and sends an error message, which the ATM displays to the customer.

- The cash dispenser module, upon approval of the withdrawal request by the ATM machine, dispenses the requested amount to the customer accurately.
- The bank server will update the customer's account balance and record the transaction details for future reference and auditing.
- Assuming the receipt printer has paper, the ATM prints out a receipt showing the date, the time, the amount withdrawn, and the remaining balance.
- Once the transaction is done, the ATM ejects the card and requests that the customer take it back to ensure the session closes securely.
- If the customer does not respond within a certain time, the ATM automatically cancels the transaction by ejecting the card for security purposes.
- In case of communication failure with the bank server or any internal malfunction, the appropriate error message is displayed on the ATM and the session safely closed down.
- It is assumed that all communications between the ATM, bank server, and account database are encrypted. Moreover, no sensitive data, like PINs, should be stored on the local memory of the ATM.
- Transaction concluded The process is regarded as complete when the customer receives cash and the system records the transaction successfully both at the ATM and bank databases.

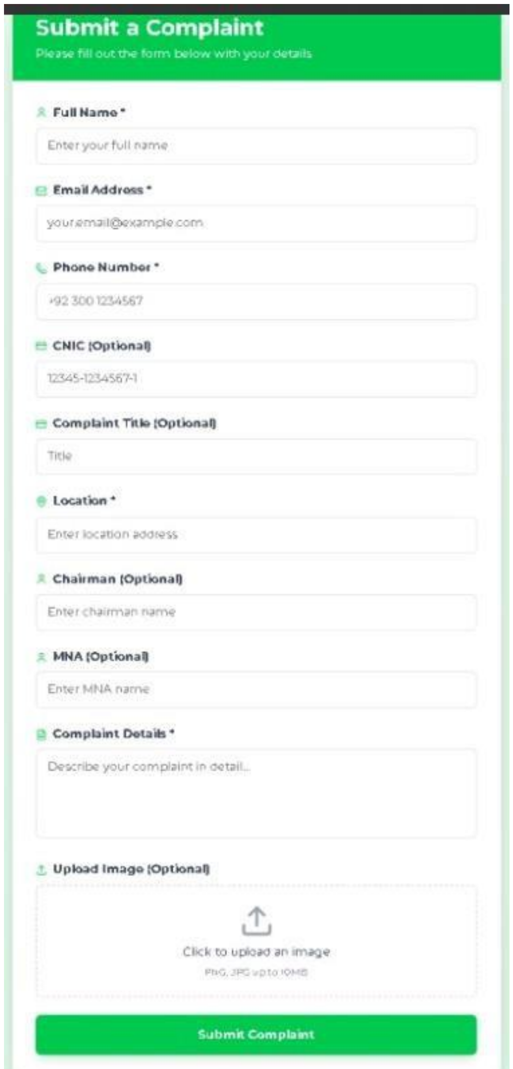


Lab Session 12

EXERCISE

Build a simple “Report a Pothole” screen:

- Create a frame (mobile layout)
- Design input fields: Address, Severity (1–10), Description, Photo Upload, Submit Button
- Apply styling, align with layout grid, and convert Submit button into a component
- Prototype a submission interaction (Attached Printout here)



Lab Session 14

EXERCISE

Build high fidelity UI/UX design of Any E-Commerce Website and attach all printouts here